CHAPTER
8 INTERIOR FINISH, DECORATIVE MATERIALS AND FURNISHINGS

General Comments

This chapter is consistent with the *International Building Code®* (IBC®), which regulates the interior finishes of buildings through the regulation of their flame spread potential. The code goes beyond interior finishes, also regulating furnishings and vegetation in buildings in certain occupancies. Additionally, the code addresses interior finishes and decorative materials in existing buildings.

This chapter is related to fire growth and spread potential in terms of the immediate effect on building occupants. The flame spread characteristics of certain materials will affect the potential fire scenarios within a building. Fire-resistance-rated construction, which is dealt with in Chapter 7 of both the IBC and this code, is more concerned with the spread of fire throughout the structure once the fire has reached a substantial size, with an emphasis on structural failure during a fire.

The regulation of flame spread can be traced back to large life-loss events, such as the Cocoanut Grove nightclub fire that killed 492 people in 1942. This fire was thought to have started when a light bulb in a basement cocktail lounge came in contact with the cotton cloth that had been applied to the ceiling for decorative purposes. Post-fire testing of the cotton cloth indicated that it had a flame spread rating of 2,500, more than 33 times the maximum allowable flame spread in today's codes. This factor, in addition to a series of problems with the egress system, led to one of the worst fire disasters in history. The need for these regulations was further emphasized after The Station nightclub fire in West Warwick, Rhode Island, where 100 people died in February 2003. The soundproofing material in the nightclub was not approved for such use and was a major factor in the fire growth.

In addition to flame spread ratings of surface materials, certain furnishing types and vegetation, such as Christmas trees, pose a significant fire hazard because of their potential fire size and intensity. The materials used in furnishings have changed dramatically from past materials and many more plastics are now used for both decoration and furnishings. Plastics not only burn more vigorously than materials like cotton and wood, but also produce more toxic fire effluents.

Purpose

The overall purpose of Chapter 8 is to regulate interior finishes, furnishings and vegetation so they do not significantly add to or create fire hazards in buildings. The provisions tend to aim at occupancies with specific risk characteristics, such as vulnerability of occupants, density of occupants, lack of familiarity with the building and societal expectations of importance. Since this is a fire code, it includes both new and existing buildings.

SECTION 801—GENERAL

801.1 Scope. The provisions of this chapter shall govern interior finish, interior trim, furniture, furnishings, decorative materials and decorative vegetation in buildings. Existing buildings shall comply with Sections 803 through 808. New buildings shall comply with Sections 804 through 808, and Section 803 of the *International Building Code.*

C This chapter reflects the same scope of issues as IBC Chapter 8 but has a slightly different emphasis. Fire codes are intended to address fire hazards of buildings and facilities over their life span; therefore, there is greater emphasis on the furnishings and contents of buildings and on the maintenance of flame spread indexes over time. Section 803 requires the same flame spread indexes as the IBC. Generally, regulating the combustibility of contents is a fairly difficult task once the building is occupied and is considered existing. Because of this difficulty, only some combustible contents and decorative materials are regulated in a limited number of occupancies. More specifically, the use of combustible furnishings, contents and decorative materials in Group A occupancies is addressed because of the high occupant load and the lack of familiarity of most occupants with the building. The type of furniture allowed in Group I occupancies is limited because of the vulnerability of the occupants and the likely fire scenarios that may occur where a building is nonsprinklered.

SECTION 802—DEFINITIONS

802.1 Definitions. The following terms are defined in Chapter 2:

FLAME SPREAD.

FLAME SPREAD INDEX.

INTERIOR FLOOR-WALL BASE.

SITE-FABRICATED STRETCH SYSTEM.

SMOKE-DEVELOPED INDEX.

C Definitions of terms can help in the understanding and application of the code requirements. This section directs the code user to Chapter 2 for the proper application of the indicated terms used in this chapter. Terms may be defined in Chapter 2 or in another International Code® (I-Code®) as indicated in Section 201.3, or the dictionary meaning may be all that is needed (see commentary, Sections 201.1 through 201.4).

SECTION 803—INTERIOR WALL AND CEILING FINISH IN EXISTING BUILDINGS

803.1 General. The provisions of this section shall limit the allowable fire performance and smoke development of interior wall and ceiling finishes in existing buildings based on location and occupancy classification. Interior wall and ceiling finishes shall be classified in accordance with Section 803 of the *International Building Code*. Such materials shall be classified in accordance with NFPA 286, as indicated in Section 803.1.1, or in accordance with ASTM E84 or UL 723, as indicated in Section 803.1.2.

Materials tested in accordance with Section 803.1.1 shall not be required to be tested in accordance with Section 803.1.2.

C This section specifically addresses existing buildings, whereas Sections 804 through 808 address both new and existing buildings. This section provides two methods for compliance with interior wall and ceiling finish fire performance requirements. Both are fire test methods that address the potential for flame to spread.

The first test is NFPA 286, which is called a "room corner test." The second test method is ASTM E84, also known as the "Steiner tunnel" or the "tunnel test." These two tests are discussed further in the commentary for Sections 803.1.1 and 803.1.2, respectively. Requirements for interior wall and ceiling finishes and interior wall and ceiling trim within existing buildings are based on occupancy. Section 803.3 and Table 803.3 address these occupancy limitations in more detail.

There are two specific exemptions to the requirements of this section listed in Sections 803.14 and 803.15. Section 803.14 exempts materials with a thickness less than 0.036 inches that are applied directly to the surface of walls and ceilings and Section 803.15 exempts the exposed portions of heavy timber construction elements that meet the requirements of the IBC for Class IV Construction.

Note that Section 803 also addresses the following:

- Textiles and expanded vinyl used as a wall or ceiling finish.
- Limitations on the use of foam.
- Polyethylene and polypropylene used as interior finish.
- Site-fabricated stretch systems.
- Facings or wood veneers intended to be applied on site over a wood substrate.
- Laminated products factory produced with an attached wood substrate.

803.1.1 Interior wall and ceiling finish materials tested in accordance with NFPA 286. Interior wall and ceiling finish materials shall be classified in accordance with NFPA 286 and tested in accordance with Section 803.1.1.1. Materials complying with Section 803.1.1.1 shall be considered to comply with the requirements of Class A specified in Section 803.1.2.

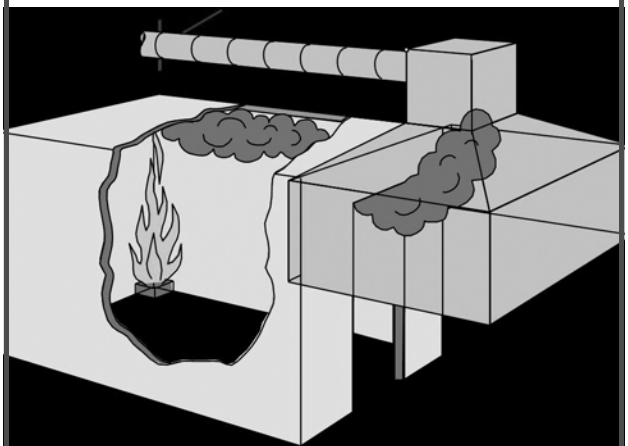
C The NFPA test method for determining compliance for interior wall and ceiling finish and trim other than textiles is found in standard 286. This test is known as a "room corner" fire test and is similar to that referenced for textile wall coverings in Section 803.5 (see NFPA 265) [see Commentary Figures 803.1.1(1) and 803.1.1(2)]. In this test, materials are mounted covering three walls of the compartment (excluding the wall containing the door) and the ceiling. In the case where testing is only for ceiling finish properties, the sample only needs to be mounted on the ceiling. Then a fire source consisting of a gas burner is placed in one corner, flush against both walls (furthest from the doorway) of the compartment with the following exposure conditions:

- 40 kilowatts (kW) for 5 minutes; then
- 160 kW for 10 minutes.

The test then measures heat release and smoke release through the collection of the fire effluents and measurement of oxygen concentrations in the exhaust duct. Heat release is calculated by the oxygen consumption principle, which has shown that heat release is a function of the decrease in oxygen concentration in the fire effluents. Thus, exhaust duct measurements include temperatures, pressures and smoke values for use in the calculations. Temperatures and heat fluxes are also measured in the room. This generally provides a more realistic understanding of the fire hazard associated with the materials (see also Section 803.5.1 for more details on the room corner test concept).

The NFPA 286 test method does not contain pass/fail criteria; however, the code provides such criteria in Section 803.1.1.1.

Commentary Figure 803.1.1(1)—Room Corner Test



Commentary Figure 803.1.1(2)—Flame in Room Corner Test

803.1.1.1 Acceptance criteria for NFPA 286. The interior finish shall comply with the following:

1. During the 40 kW exposure, flames shall not spread to the ceiling.
2. The flame shall not spread to the outer extremity of the sample on any wall or ceiling.
3. Flashover, as defined in NFPA 286, shall not occur.
4. The peak heat release rate throughout the test shall not exceed 800 kW.
5. The total smoke released throughout the test shall not exceed 1,000 m^2.

C As noted in Section 803.1.1, there are two levels of exposure during an NFPA 286 fire test in order to better represent a growing fire: 40 kW fire size for 5 minutes and 160 kW for 10 minutes. The 40 kW exposure represents the beginning of a fire where the initial spread is critical; therefore, the stated criteria is that the fire cannot spread to the ceiling. The 160 kW exposure is obviously a more intense fire situation and the criteria relates to preventing flashover (as defined by NFPA 286) and the extent of flame spread throughout the entire test assembly. There is also a total smoke production criterion of 1,000 m^2. It should be noted that the criteria used in NFPA 286 and NFPA 265 to determine if flashover has occurred would include any two of the following:

- Heat release exceeds 1 mega watt (MW).
- Heat flux at the floor exceeds 20 kW/m^2.
- Average upper layer temperature exceeds 1112°F (600°C).
- Flames exit the doorway.
- Autoignition of paper target on the floor occurs.

Furthermore, it should be noted that there is an additional criterion when applying NFPA 286 for new buildings, which is a maximum peak heat release rate of 800 kW. The reasoning for this criterion relates to the fact that some poorer-performing materials can achieve compliance with the flashover criteria but could have a higher peak heat release rate.

803.1.2 Interior wall and ceiling finish materials tested in accordance with ASTM E84 or UL 723. Interior wall and ceiling finishes shall be classified in accordance with ASTM E84 or UL 723. Such interior finish materials shall be grouped in the following classes in accordance with their flame spread and smoke-developed indices:

Class A: Flame spread index 0–25; smoke-developed index 0–450.

Class B: Flame spread index 26–75; smoke-developed index 0–450.

Class C: Flame spread index 76–200; smoke-developed index 0–450.

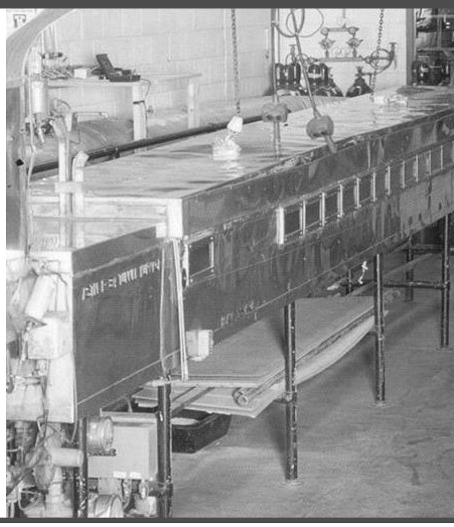
Exception: Materials tested in accordance with Section 803.1.1 and as indicated in Sections 803.1.3 through 803.15.

C Wall and ceiling interior finish and trim materials are required to have limits on flame spread and smoke-developed indexes as prescribed in Section 803.3, based on occupancy. ASTM E84 is one of the tests available to demonstrate compliance with the requirements of Section 803. This test method has been around since 1944 [see Commentary Figure 803.1.2(1)]. It is the primary method used but, as noted earlier, NFPA 286 is an alternative test, which is discussed in the commentary for Section 803.1.1.

ASTM E84 is intended to determine the relative burning behavior of materials on exposed surfaces, such as ceilings and walls, by visually observing the flame spread along the test specimen [see Commentary Figure 803.1.2(2)]. Flame spread and smoke density indexes are then reported. The test method is not appropriate for materials that are not capable of supporting themselves, or of being supported in the test tunnel. There may also be concerns with materials that drip, melt or delaminate and that are very thin. A distinction is made, therefore, for textile wall or ceiling coverings, expanded vinyl wall or ceiling coverings, wood veneers applied on site, foam plastic insulation materials, high density polyethene and polypropylene and site-fabricated stretch systems (see Sections 803.5 through 803.10).

ASTM E84 establishes a flame spread index based on the area under a curve when the actual flame spread distance is plotted as a function of time. The code has divided the acceptable range of flame spread indexes (0–200) into three classes: Class A (0–25), Class B (26–75) and Class C (76–200). For all three classes, the code has established a common acceptable range of smoke-developed index of 0–450. An indication of relative fire performance is as follows: an inorganic reinforced-cement board has a flame spread and smoke-developed index of zero, while select grade red oak wood flooring has a flame spread and smoke-developed index of 100. Not to preclude more detailed information resulting from an ASTM E84 test report, Commentary Figure 803.1.2(3) identifies the typical flame spread properties of certain building materials.

Commentary Figure 803.1.2(1)—ASTM E84 Tunnel Test



Commentary Figure 803.1.2(2)—Flame in Tunnel Test

Commentary Figure 803.1.2(3)—Typical Flame Spread of Common Materials

Material	Flame spread
Glass-fiber sound-absorbing blanks	15 to 30
Mineral-fiber sound-absorbing panels	10 to 25
Shredded wood fiberboard (treated)	20 to 25
Sprayed cellulose fibers (treated)	20
Aluminum (with baked enamel finish on one side)	5 to 10
Asbestos-cement board	0
Brick or concrete block	0
Cork	175
Gypsum board (with paper surface on both sides)	10 to 25
Northern pine (treated)	20
Southern pine (untreated)	130 to 190
Plywood paneling (untreated)	75 to 275
Plywood paneling (treated)	100
Carpeting	10 to 600
Concrete	0

803.1.3 Interior wall and ceiling finish materials with specific requirements. The materials indicated in Sections 803.4 through 803.15 shall be tested as indicated in the corresponding sections.

C This section states that the purpose of Sections 803.4 through 803.15 is to provide specific testing requirements for the interior wall and ceiling finish materials that are listed in each individual section. The materials that are included are: fire-retardant coatings, textile wall and ceiling coverings, expanded vinyl wall and ceiling coverings, high-density polyethylene (HDPE) and polypropylene (PP), site-fabricated stretch systems, foam plastic materials, facings or wood veneers intended to be applied on site over a wood substrate, and laminated products factory produced with an attached wood substrate. There are also testing exemptions for thin materials and exposed portions of heavy timber construction.

803.2 Stability. Interior finish materials regulated by this chapter shall be applied or otherwise fastened in such a manner that such materials will not readily become detached where subjected to room temperatures of 200°F (93°C) for not less than 30 minutes.

C Interior finishes are not to become detached for a minimum of 30 minutes under exposure to elevated temperatures [200°F (93°C)]. No standard test method has yet been developed to evaluate this requirement. Some sections of the IBC, however, do offer some additional guidance. For example, the performance of the method of attachment of finish materials during a fire-resistance test will usually be an adequate indication of performance. The stability criterion is necessary because, if these materials were to fall off walls or the ceiling during a fire, they may contribute to the fire and increase the hazard beyond what is typically expected.

803.3 Interior finish requirements based on occupancy. Interior wall and ceiling finish shall have a flame spread index not greater than that specified in Table 803.3 for the group and location designated. Interior wall and ceiling finish materials tested in accordance with NFPA 286, and meeting the acceptance criteria of Section 803.1.1.1, shall be used where a Class A classification in accordance with ASTM E84 or UL 723 is required.

C The requirements for flame spread indexes for interior finish materials applied to walls and ceilings are contained in Table 803.3. The referenced test for determining flame spread indexes is ASTM E84, which establishes a relative measurement of flame spread across the surface of the material. The classifications used in Table 803.3 are defined in Section 803.1.2 (see the commentary to Section 803.1.2 for additional information on the uses and limitations of the test procedure). Again, NFPA 286 can be used as an alternative to ASTM E84. Passing NFPA 286 means that the material would be considered to be equivalent to a Class A material.

TABLE 803.3—INTERIOR WALL AND CEILING FINISH REQUIREMENTS BY OCCUPANCY[k]

GROUP	SPRINKLERED[l]			NONSPRINKLERED		
	Interior exit stairways and ramps and exit passageways[a, b]	Corridors and enclosure for exit access stairways and ramps	Rooms and enclosed spaces[c]	Interior exit stairways and ramps and exit passageways[a, b]	Corridors and enclosure for exit access stairways and ramps	Rooms and enclosed spaces[c]
A-1 and A-2	B	B	C	A	A[d]	B[e]
A-3[f], A-4, A-5	B	B	C	A	A[d]	C
B, E, M, R-1, R-4	B	C[m]	C	A	B[m]	C
F	C	C	C	B	C	C
H	B	B	C[g]	A	A	B
I-1	B	C	C	A	B	B
I-2	B	B	B[h, i]	A	A	B
I-3	A	A[j]	C	A	A	B
I-4	B	B	B[h, i]	A	A	B
R-2	C	C	C	B	B	C
R-3	C	C	C	C	C	C
S	C	C	C	B	B	C
U	No Restrictions			No Restrictions		

For SI: 1 inch = 25.4 mm, 1 square foot = 0.0929 m^2.

a. Class C interior finish materials shall be allowed for wainscoting or paneling of not more than 1,000 square feet of applied surface area in the grade lobby where applied directly to a noncombustible base or over furring strips applied to a noncombustible base and fireblocked as required by Section 803.11 of the *International Building Code*.

b. In exit enclosures of buildings less than three stories in height of other than Group I-3, Class B interior finish for nonsprinklered buildings and Class C for sprinklered buildings shall be permitted.

TABLE 803.3—INTERIOR WALL AND CEILING FINISH REQUIREMENTS BY OCCUPANCY[k]—continued

c. Requirements for rooms and enclosed spaces shall be based on spaces enclosed by partitions. Where a fire-resistance rating is required for structural elements, the enclosing partitions shall extend from the floor to the ceiling. Partitions that do not comply with this shall be considered as enclosing spaces and the rooms or spaces on both sides shall be considered as one. In determining the applicable requirements for rooms and enclosed spaces, the specific occupancy thereof shall be the governing factor, regardless of the group classification of the building or structure.
d. Lobby areas in Group A-1, A-2 and A-3 occupancies shall be not less than Class B materials.
e. Class C interior finish materials shall be allowed in Group A occupancies with an occupant load of 300 persons or less.
f. In places of religious worship, wood used for ornamental purposes, trusses, paneling or chancel furnishing shall be allowed.
g. Class B material is required where the building exceeds two stories.
h. Class C interior finish materials shall be allowed in administrative spaces.
i. Class C interior finish materials shall be allowed in rooms with a capacity of four persons or less.
j. Class B materials shall be allowed as wainscoting extending not more than 48 inches above the finished floor in corridors.
k. Finish materials as provided for in other sections of this code.
l. Applies where the vertical exits, exit passageways, corridors or rooms and spaces are protected by an approved automatic sprinkler system installed in accordance with Section 903.3.1.1 or 903.3.1.2.
m. Corridors in ambulatory care facilities shall have a Class B or better interior finish material.

C This table prescribes the minimum requirements for interior finishes applied to walls and ceilings; therefore, the use of a Class A material in an area that requires a minimum Class B material is always allowed. Likewise, when the table requires Class C materials, Classes A and B can be used. The requirements are based on the use of the space. To determine the applicable criteria, first determine whether the space is an exit passageway, interior exit stairway or interior exit ramp; a corridor; or a room or enclosed space. Interior finishes in spaces that are not separated from a corridor (for example, waiting areas in business or health care facilities) must comply with the requirements for a corridor space. As shown in the table, the code places a higher emphasis on the allowable flame spread index for exits than for enclosed rooms because of the critical nature and relative importance of maintaining exit integrity to aid in building evacuation. Numerous notes amend the basic requirements of Table 803.3. Notes a through i apply only where specifically referenced in the table.

Note a

A limited amount of combustible wainscotting or other paneling material does not appreciably reduce the level of safety of an exit element; therefore, up to 1,000 square feet (93 m^2) of Class C wainscotting or paneling is permitted by Note a in a grade-level lobby used as an exit element when applied in accordance with IBC Section 803.3.

Note b

This note allows the reduction of flame spread indexes in exit enclosures in buildings less than three stories high, not including Group I-3 occupancies. The time required to exit a building less than three stories is much shorter than that for taller buildings and enough time usually exists to use other exit enclosures.

Note c

Because the intended use of certain rooms in buildings is sometimes more hazardous than the group classification for the overall building, the finish classification must be determined by the group and occupancy classification for the room or area. Additionally, rooms or enclosed spaces not properly separated from one another must be looked at as a single space for the purposes of applying flame spread index requirements.

Note d

This note allows lobbies in Group A-1, A-2 and A-3 occupancies to be Class B instead of Class A. This is likely a result of the low fire load in such areas.

Note e

Note e recognizes that with a relatively small number of occupants, egress is accomplished more quickly and the activities tend to be more structured and manageable than with a large number of occupants. When the design occupant load is 300 or less in rooms or spaces of Group A-1 and A-2 occupancies, the interior finish materials may be Class C instead of Class B for rooms and enclosed spaces.

Note f

This note correlates with the allowance in Section 807.5.1.3 for places of religious worship.

Note g

The time required for exiting increases with the number of stories of the structure. Interior finish materials can play a major role in fire spread within a structure. Because of the materials stored and used within a Group H occupancy, interior wall and ceiling finish materials with high heat release (or flame spread) potential can have a significant impact on the way a fire spreads throughout the building. To abate the hazard of rapid fire growth, Note g places a further restriction of a minimum of a Class B classification on interior finishes of enclosed spaces within sprinklered buildings when the building exceeds two stories.

Note h

Spaces that are used as offices in Group I-2 and I-4 occupancies have a low occupant load, the activity in those spaces is generally not very hazardous and the occupants are not as vulnerable as the general population of those facilities, partially because of their knowledge of the exits; therefore, Class C materials are allowed in administrative spaces.

Note i

Rooms in Group I-2 and I-4 occupancies with low occupant loads (four or less) pose a low risk and are quickly evacuated; therefore, Class C finish materials are appropriate.

Note j

In Group I-3 occupancy corridors, Note j allows interior finish materials, such as wainscotting, that are a maximum height of 48 inches (1219 mm) above the floor to be Class B. This reduction from Class A is based on full-scale fire research demonstrating that Class B wall finish used on the lower 4 feet (1219 mm) of a corridor wall is not likely to spread fire because fire primarily spreads on ceilings and upper walls before affecting the lower portions of walls.

Note k

This note is referenced in the title of Table 803.3 and refers to other sections of the code that may have further restrictions or allowances for material classification for specific uses.

Note l

Sprinklered facilities provide more protection to occupants than nonsprinklered facilities. In many instances, materials with lower restrictions on fire performance may be used, as shown in Table 803.3. The footnote clarifies what areas of the building need to be sprinklered in order to qualify for these reductions in interior finish classifications.

Note m

The Class B requirement for corridors in ambulatory facilities is consistent with federal regulations for buildings that are provided with an automatic sprinkler system. It is also shown in the column for the buildings without an automatic sprinkler system to prevent confusion between the two types.

803.4 Fire-retardant coatings. The required flame spread or smoke-developed index of surfaces in existing buildings shall be allowed to be achieved by application of *approved* fire-retardant coatings, paints or solutions to surfaces having a flame spread index exceeding that allowed. Such applications shall comply with NFPA 703 and the required fire-retardant properties shall be maintained or renewed in accordance with the manufacturer's instructions. The fire-retardant paint, coating or solution shall have been assessed by testing over the same substrate to be used in the application.

C Many times, fire retardants can be used to reduce the flame spread index of a material that, without treatment, has an index higher than permissible. It should be recognized that the most desirable situation is to have a material that has a specific flame spread rating or has been tested and passed in accordance with NFPA 286. It is also recognized that in existing buildings there are often situations where such an approach is not possible and the application of fire retardants is the only solution. Flame retardants may either be factory or field applied. A wide variety of proprietary flame retardants for field application are on the market. Both factory- and field-applied products may require reapplication to sustain the flame retardancy of a particular article. When reviewing an installation, the fire code official should require the owner or other responsible person to submit a manufacturer's or contractor's certificate documenting a given treatment. This documentation should specify if a particular treatment must be repeated and, if so, at what intervals. If no retreatment interval is specified, the manufacturer's or contractor's warranty should be recognized as the retreatment interval because it is the longest period for which that individual accepts responsibility for the product's performance. Fire code officials evaluating flame-retardant treatments should carefully evaluate the performance conditions of these preparations. Many of them are not intended to be exposed to direct sunlight, high humidity or weather. Flame-retardant treatment must be used or applied in a manner consistent with the manufacturer's instructions.

There are several methods of increasing the flame resistance of materials. The strategy is either to slow the ignition process or control combustion itself. Commentary Figure 803.4 lists various methods and whether they can be achieved in the field.

It should also be noted that in more recent times there has been growing concern over the use of polybrominated diphenyl ethers (PBDE) to achieve flame resistance due to the possible health effects. Some states, including California, have passed legislation to ban the use of such products. Several manufacturers have also begun a voluntary phase-out of such chemicals.

Commentary Figure 803.4—Applied Flame-Retardant Coating

METHOD OF ACHIEVING FLAME RESISTANCE	APPROPRIATE USE	APPLICATION	OTHER INFORMATION
Chemical	Synthetics; Plastics	Chemical added during manufacturing process Example: Polymers	Actual chemical change occurs Changes the behavior of materials
Impregnation	Absorbent or porous materials; not wood (too dense)	In the field by spray or immersion	Sometimes done at wet pulp stage (Paper, acoustical tile and building panels)
Coating	Nonabsorbant materials Example: Wood paneling, ceiling tiles, etc.	In the field by spray, brush or roller application	Actively inhibits flame spread or simply provides a noncombustible surface Needs periodic renewal per manufacturer
Pressure Impregnation	Dense, nonabsorbent materials such as wood	Only during manufacturing	Replaces air pockets with fire-retardant solution Chemical deposits while drying Vacuum-pressure methods are used to impregnate materials Far more effective than impregnation alone

803.5 Textile wall coverings. Where used as interior wall finish materials, textile wall coverings, including materials having a woven, nonwoven, napped, tufted, looped or similar surface, shall be tested in the manner intended for use, using the product mounting system, including adhesive, and shall comply with the requirements of Section 803.1.1, 803.5.1 or 803.5.2.

C This section is primarily intended to apply to carpet and carpet-like wall coverings that include textile materials having woven or nonwoven, napped, tufted, looped or similar surfaces. If not addressed, these easily ignitable materials can contribute extensively to a fire. There are three testing options for these coverings as given in Section 803.1.1, 803.5.1 or 803.5.2. In each case, it shall be tested in the manner intended for use, using the product mounting system, including adhesive.

803.5.1 Room corner test for textile wall coverings and expanded vinyl wall coverings. Textile wall coverings and expanded vinyl wall coverings shall meet the criteria of Section 803.5.1.1 when tested in the manner intended for use in accordance with the Method B protocol of NFPA 265 using the product mounting system, including adhesive. Test specimen preparation and mounting shall be in accordance with ASTM E2404.

C As an alternative to a Class A flame spread index and sprinklers, testing in accordance with NFPA 265 may be used for textile wall coverings. Just as discussed for NFPA 286, NFPA 265 is known as a full-scale room corner fire test. This test helps to determine the contribution of textile wall coverings to overall fire growth and spread in a compartment fire. Past research conducted with this kind of configuration has shown that flame spread indexes produced by ASTM E84 or UL 723 may not reliably predict the fire behavior of textile wall coverings in realistic fire scenarios. Thus, NFPA 265 is based on a more reliable test procedure, developed at the University of California. The research findings are described in a report from the University of California Fire Research Laboratory titled, “Room Fire Experiments of Textile Wall Coverings.” The NFPA 265 test is only slightly different from NFPA 286. NFPA 286 is more severe in three ways:

1. The gas diffusion burner used to expose the material on the wall in the room fire test starts with a heat release rate exposure of 40 kW for the first 5 minutes in both tests, but it is then followed by 150 kW for 10 minutes in NFPA 265 and 160 kW for 10 minutes in NFPA 286.
2. The gas burner is placed 2 inches (51 mm) from each of the walls in NFPA 265, whereas it is placed flush against both walls in NFPA 286.
3. The test sample is mounted on the walls only in NFPA 265, whereas it is mounted on both the walls and ceiling in NFPA 286.

A key result of the difference in intensity and location of the gas burner is that the burner flame does not reach the ceiling during the 150 kW exposure (while it does reach the ceiling during the 160 kW exposure in NFPA 286). Therefore, NFPA 265 is not considered suitable for testing ceiling coverings.

It should be noted that the code does not require measurement of smoke release from materials tested to NFPA 265 unless the material is newly introduced but does have a smoke pass/fail criterion for NFPA 286. NFPA 265 has two test methods, but the code only allows the use of Method B. In the Method A test protocol, 2-foot-wide (610 mm) strips of the material are mounted on the two walls closest to the corner with the burner, whereas in the Method B test protocol the sample is mounted completely covering three walls (except for the wall containing the door). Therefore, Method B is more severe. This test method does not contain pass/fail criteria, just as NFPA 286 does not. The code therefore provides such criteria, based on the use of only the Method B test protocol from NFPA 265, in Section 803.5.1.1. It should be noted that the IBC also only allows the use of Method B from NFPA 265. The code applies to existing buildings and the IBC would only apply to new buildings with regard to this particular test.

803.5.1.1 Acceptance criteria for NFPA 265 Method B test protocol. Where testing to NFPA 265, the interior finish shall comply with the following:

1. During the 40-kW exposure, flames shall not spread to the ceiling.
2. The flame shall not spread to the outer extremities of the samples on the 8-foot by 12-foot (203 by 305 mm) walls.
3. Flashover, as defined in NFPA 265, shall not occur.
4. For newly introduced wall coverings, the total smoke released throughout the test shall not exceed 1,000 m^2.

C The criteria are very similar to those used in NFPA 286. The pass/fail criteria in the code are as follows:

The flame cannot spread to the ceiling when the textile is exposed to the burner at 40 kW. With the burner at 150 kW, the following criteria must be met:

- The fire cannot reach the outer areas of the 8-foot by 12-foot (2438 mm by 8657 mm) wall.
- Flashover, as defined by NFPA 265, must not occur.
- Newly introduced wall coverings are limited to a total smoke release of 1000 m^2 (10,760 ft^2).

803.5.2 Acceptance criteria for wall and ceiling coverings. Textile wall and ceiling coverings shall have a Class A flame spread index in accordance with ASTM E84 or UL 723, and be protected by an *automatic sprinkler system* installed in accordance with Section 903.3.1.1 or 903.3.1.2. Test specimen preparation and mounting shall be in accordance with ASTM E2404.

C This section requires that textile materials on walls or ceilings have a Class A flame spread index and be located in a sprinklered area in accordance with NFPA 13 or 13R. As an alternative to a Class A flame spread index and sprinklers, testing in accordance with NFPA 265 may be used for textile wall coverings. Just as discussed for NFPA 286, NFPA 265 is known as a full-scale room corner fire test. This test helps to determine the contribution of textile wall coverings to overall fire growth and spread in a compartment fire. Past research conducted with this kind of configuration has shown that flame spread indexes produced by ASTM E84 or UL 723 may not reliably predict the fire behavior of textile wall coverings in realistic fire scenarios. Thus, NFPA 265 is based on a more reliable test procedure, developed at the University of California. The research findings are described in a report from the University of California Fire Research Laboratory titled, “Room Fire Experiments of Textile Wall Coverings.” The NFPA 265 test is only slightly different from NFPA 286. NFPA 286 is more severe in three ways:

1. The gas diffusion burner used to expose the material on the wall in the room fire test starts with a heat release rate exposure of 40 kW for the first 5 minutes in both tests, but it is then followed by 150 kW for 10 minutes in NFPA 265 and 160 kW for 10 minutes in NFPA 286.
2. The gas burner is placed 2 inches (51 mm) from each of the walls in NFPA 265, whereas it is placed flush against both walls in NFPA 286.
3. The test sample is mounted on the walls only in NFPA 265, whereas it is mounted on both the walls and ceiling in NFPA 286.

A key result of the difference in intensity and location of the gas burner is that the burner flame does not reach the ceiling during the 150 kW exposure (while it does reach the ceiling during the 160 kW exposure in NFPA 286). Therefore, NFPA 265 is not considered suitable for testing ceiling coverings.

It should be noted that the code does not require measurement of smoke release from materials tested to NFPA 265 unless the material is newly introduced but does have a smoke pass/fail criterion for NFPA 286. NFPA 265 has two test methods, but the code only allows the use of Method B. In the Method A test protocol, 2-foot-wide (610 mm) strips of the material are mounted on the two walls closest to the corner with the burner, whereas in the Method B test protocol the sample is mounted completely covering three walls (except for the wall containing the door). Therefore, Method B is more severe. This test method does not contain pass/fail criteria, just as NFPA 286 does not. The code therefore provides such criteria, based on the use of only the Method B test protocol from NFPA 265, in Section 803.5.1.1. It should be noted that the IBC also only allows the use of Method B from NFPA 265. The code applies to existing buildings and the IBC would only apply to new buildings with regard to this particular test.

803.6 Textile ceiling coverings. Where used as interior ceiling finish materials, textile ceiling coverings, including materials having a woven, nonwoven, napped, tufted, looped or similar surface and carpet or similar textile materials, shall be tested in the manner intended for use, using the product mounting system, including adhesive, and shall comply with the requirements of Section 803.1.1 or 803.5.2.

C This section is basically the same as Section 803.5.1 with one exception focused on preparing test specimens for the tunnel test (ASTM E84). The reason for this difference is that existing materials would possibly be out of compliance since they were tested prior to the development of ASTM E2404, which has specific mounting requirements for materials during the test. It was felt reasonable in existing buildings to hold new textile wall and ceiling coverings to this higher standard. The correct specimen preparation and mounting method for textile, paper and vinyl wall and ceiling coverings tested in accordance with the ASTM E84 (Steiner tunnel) test is ASTM E2404. The ASTM E05 Committee on Fire Standards developed a standard practice for test specimen preparation and mounting, ASTM E2404, for textile, paper or vinyl wall or ceiling coverings specifically to provide a mandatory, standardized way of preparing test specimens and mounting them in the tunnel. This replaced optional guidance on mounting methods found in the appendix of ASTM E84 and ensures testing consistency.

This section also allows the use of NFPA 265 and NFPA 286 with the same criteria as cited in Section 803.5.1. Note that NFPA 265 is not applicable for the testing of ceiling materials due to the limitations of that test (see commentary, Section 803.5.1).

803.7 Expanded vinyl wall coverings. Where used as interior wall finish materials, expanded vinyl wall coverings shall be tested in the manner intended for use, using the product mounting system, including adhesive, and shall comply with the requirements of Section 803.1.1, 803.5.1 or 803.5.2.

C This section is specific to the use of expanded vinyl as wall coverings. It requires compliance with one of three testing methods. These methods are: NFPA 286 as listed in Section 803.1.1, NFPA 265 Method B as listed in Section 803.5.1 or ASTM E84 as listed in Section 803.5.2. The coverings are required to be tested as they are intended to be used, including mounting systems and adhesives.

This section does not limit the requirements to newly introduced materials but instead applies to any expanded vinyl wall coverings. See also the commentaries to Sections 803.5.2 and 803.5.1.

803.8 Expanded vinyl ceiling coverings. Where used as interior ceiling finish materials, expanded vinyl ceiling coverings shall be tested in the manner intended for use, using the product mounting system, including adhesive, and shall comply with the requirements of Section 803.1.1 or 803.5.2.

C This section is specific to the use of expanded vinyl as ceiling coverings. It requires compliance with one of two testing methods. These methods are: NFPA 286, as listed in Section 803.1.1 or ASTM E84, as listed in Section 803.5.2. It is required that the specific testing of the coverings be performed in the exact manner that the product is intended for use, using the actual mounting system, and includes the adhesive.

This section does not limit the requirements to newly introduced materials but instead applies to any expanded vinyl ceiling coverings. See also the commentary to Section 803.5.2.

[BF] 803.9 High-density polyethylene (HDPE) and polypropylene (PP). Where high-density polyethylene or polypropylene is used as an interior finish, it shall comply with Section 803.1.1.

C High-density polyethylene (HDPE) and polypropylene (PP) are thermoplastics that give off considerable energy and produce a pooling flammable liquid fire when they burn. Recent full-scale room-corner tests of HDPE using NFPA 286 have demonstrated a significant hazard. These tests had to be terminated prior to the standard 15-minute duration due to flashover occurring while there was still much of the product left to burn. Extensive flammable liquid pool fires occurred during the tests, yet this same material, when tested in accordance with the tunnel test, ASTM E84, is often given a flame spread index of 25 or less. However, the resulting test is so intense some labs will not test HDPE partitions due to the damage such tests can do to the tunnel. This section ensures that when using HDPE partitions, they will be formulated in such a manner as to reduce the hazard that they present by specifically requiring compliance with NFPA 286. The following are some of the data gleaned from one of the NFPA 286 tests: Peak Heat Release Rate 1733 kW; Total Heat Released 121 MJ; Peak Heat Flux to the Floor 35.2 kW/m^2; Peak Average Ceiling Temperature 805°C, 1481°F. Polypropylene is very similar in material and fire performance.

[BF] 803.10 Site-fabricated stretch systems. Where used as newly installed interior wall or interior ceiling finish materials, site-fabricated stretch systems containing all three components described in the definition in Chapter 2 shall be tested in the manner intended for use, and shall comply with the requirements of Section 803.1.1 or 803.1.2. If the materials are tested in accordance with ASTM E84 or UL 723, specimen preparation and mounting shall be in accordance with ASTM E2573.

C Site-fabricated stretch systems are interior finish materials that are stretched taut across walls or ceilings with a frame that holds a fabric and core. These systems are now being used extensively because they can stretch to cover decorative walls and ceilings with unusual looks and shapes. The systems consist of three parts: a fabric (or vinyl), a frame and an infill core material. Therefore, they must be fire tested like all other interior wall and ceiling materials, using either ASTM E84 (the

Steiner tunnel test) or NFPA 286 (the room-corner test). The ASTM Committee on Fire Standards (ASTM E05) has issued ASTM E2573, a standard practice for specimen preparation and mounting of site-fabricated stretch systems. It is not a test method but rather a mounting method that was developed specifically for use when testing these materials in the ASTM E84 Steiner tunnel test. Before it was issued, there was no correct mandatory way to test these systems. The testing has often been done of each component separately instead of testing the composite system. That is an inappropriate way to test and not the safe way to conduct the testing. Now that a consensus standard method of testing exists, the code recognizes it.

It is important to note that these materials are not curtains or drapes because they are not free hanging like curtains. Therefore, it would be inappropriate for them to be tested using NFPA 701, a test for vertically hanging fabrics, or any other test that was developed for free-hanging materials.

As previously stated, it is important to test all components together. If the system in question does not contain all three components of site-fabricated stretch systems, then such systems should be tested in a manner appropriate to their use. In particular, systems that contain a stretch membrane only and no core material have been shown to behave very differently in a fire situation than site-fabricated stretch systems. It is important that the correct mounting method be used for each system.

803.11 Foam plastic materials. Foam plastic materials shall not be used as interior wall and ceiling finish unless specifically allowed by Section 803.11.1 or 803.11.2. Foam plastic materials shall not be used as interior trim unless specifically allowed by Section 804.2.

C This section allows foam plastic interior wall and ceiling finish materials to be used in accordance with Section 803.11.1 or 803.11.2. The ASTM E84 test has the potential to produce misleading fire test results when used to test foam plastic interior wall and ceiling finish materials. In particular, if the foam plastics have low density, the amount of material may be insufficient for the ASTM E84 Steiner tunnel test to properly evaluate the plastics' flame spread characteristics. Moreover, if the foams are thermoplastic, they are likely to melt and fall away from the flame front when exposed in the test and provide artificially low flame spread information on the surface of the test sample, and thus give an inaccurate indication of the fire hazard. It was thus generally agreed, in the early 1970s, that foam insulation products should be assessed for combustibility with a test that represents more realistically the way the materials behave in actual applications, and tests, such as UL 1715, UL 1040, FM 4880 and NFPA 286, were developed for such purposes to be used instead of ASTM E84.

803.11.1 Foam plastics. Foam plastics shall be allowed to be used as interior wall and ceiling finish only where in accordance with Section 2603.9 of the *International Building Code*. This section shall apply both to exposed foam plastics and to foam plastics used in conjunction with a textile or vinyl facing or cover.

C This section requires that where foam plastics are used as interior wall or ceiling finish, they must meet the requirements of IBC Section 2603.9. The IBC states that foam plastic is not permitted to be used exposed as interior finish unless it has been tested under actual fire conditions. The IBC requires foam plastic insulation to be tested in accordance with a standard such as FM 4880, UL 1040, NFPA 286 or UL 1715, all of which are large-scale fire tests measuring heat release. NFPA 286 does not have pass/fail criteria; those criteria are contained in Section 803.1.1 of the code. This section also clarifies that covering the foam plastic with a vinyl or textile cover (as is often done in high school gymnasiums) will not change the requirements, so the same tests should be applied to the covered foam.

803.11.2 Thermal barrier for foam plastics. Foam plastic material shall be allowed if it is separated from the interior of the building by a thermal barrier in accordance with Section 2603.4 of the *International Building Code.*

C Another option the code allows is to use foam plastic where it is separated from the room interior by means of a thermal barrier, so that the foam plastic is not exposed and is not actually used as interior finish. This section refers the user to IBC Section 2603.4 to address how this is to be accomplished. IBC Section 2603.4 requires a 1/2-inch (12.7 mm) gypsum wallboard or equivalent thermal barrier. IBC Section 2603.4 also provides criteria for testing equivalent thermal barriers.

803.12 Facings or wood veneers intended to be applied on site over a wood substrate. Facings or veneers intended to be applied on site over a wood substrate shall comply with one of the following:

1. The facing or veneer shall meet the criteria of Section 803.1.1 when tested in accordance with NFPA 286 using the product mounting system, including adhesive, described in Section 5.8.9 of NFPA 286.
2. The facing or veneer shall have a Class A, B or C flame spread index and smoke-developed index based on the requirements of Table 803.3, in accordance with ASTM E84 or UL 723. Test specimen preparation and mounting shall be in accordance with ASTM E2404.

C This section details the proper way to test facings or wood veneers intended to be applied over a wood substrate. They are to be treated the same way as any other wall or ceiling covering applied "on site" to a wood substrate, but differently from panels where the facing or veneer is applied in the factory over the wood substrate and the entire panel is installed. The following section addresses this situation. This section allows the use of either NFPA 286 or ASTM E84. As noted, the ASTM E2404 mounting method must be applied if the ASTM E84 test method is chosen.

803.13 Laminated products factory produced with an attached wood substrate. Laminated products factory produced with an attached wood substrate shall comply with one of the following:

1. The laminated product shall meet the criteria of Section 803.1.1 when tested in accordance with NFPA 286 using the product mounting system, including adhesive, of actual use.
2. The laminated product shall have a Class A, B or C flame spread index and smoke-developed index based on the requirements of Table 803.3, in accordance with ASTM E84 or UL 723. Test specimen preparation and mounting shall be in accordance with ASTM E2579.

C This section provides the product testing requirements for the use of laminated products with an attached wood substrate as interior wall or ceiling finishes. This section allows the use of either NFPA 286 or ASTM E84. As noted, the ASTM E2579 mounting method must be applied if the ASTM E84 test method is chosen.

803.14 Thickness exemption. Materials having a thickness less than 0.036 inch (0.9 mm) applied directly to the surface of walls or ceilings shall not be required to be tested.

C This section provides a specific testing exemption for thin materials that are directly applied to the surfaces of walls or ceilings. It is not necessary for these thin materials to be tested due to the negligible effect they have on the overall wall or ceiling construction fire performance.

803.15 Heavy timber exemption. Exposed portions of building elements complying with the requirements of Type IV construction in accordance with the *International Building Code* shall not be subject to interior finish requirements.

C This section provides a specific application interior finish requirement exemption for the use of heavy timber exposed construction building elements that meet the IBC requirements for Type IV construction. Thus, it is not necessary to meet the interior finish requirements of this section if the requirements for Type IV construction in the IBC are met.

SECTION 804—INTERIOR WALL AND CEILING TRIM AND INTERIOR FLOOR FINISH IN NEW AND EXISTING BUILDINGS

804.1 Interior trim. Combustible trim in new and existing buildings, excluding *handrails* and guards, shall not exceed 10 percent of the specific wall or ceiling areas to which it is attached. Other than foam plastic, material used as interior trim shall comply with Section 804.1.1 or 804.1.2. Foam plastic used as interior trim shall comply with Section 804.2.

C In accordance with Section 801.1, Section 804.1 addresses both new and existing buildings whereas Section 803 is applicable only to existing buildings. In occupancies of any group, unless otherwise noted in the code, all combustible material used as interior trim, with the exception of foam plastic, is to meet the testing requirements of either Section 804.1.1 or 804.1.2. Only one testing method is required.

804.1.1 Testing in accordance with NFPA 286. Interior trim material shall be tested in accordance with NFPA 286 and comply with the acceptance criteria in Section 803.1.1.1. Where the interior trim material has been tested as an interior finish in accordance with NFPA 286 and complies with the acceptance criteria in Section 803.1.1.1, it shall not be required to be tested for flame spread index and smoke-developed index in accordance with ASTM E84 or UL 723.

C This section lists the first testing option for combustible material used as interior trim as NFPA 286 and provides the section reference for the acceptance criteria. Chapter 8 of the IBC and Section 803.1.1 of this code already make it clear that any material that meets the criteria of Section 803.1.1.1 is permitted to be used for interior finish. If the material is allowed to be used to cover the entire wall or ceiling, it is also allowed to be used to cover 10 percent of it.

804.1.2 Testing in accordance with ASTM E84 or UL 723. Material, other than foam plastic, used as interior trim shall have minimum Class C flame spread and smoke-developed indices, when tested in accordance with ASTM E84 or UL 723, as described in Section 803.1.2.

C This section lists the second testing option for combustible material used as interior trim as ASTM E84 or UL 723, with the result that the minimum classification of all trim must be at least Class C. The criteria for interior trim (whether a foam plastic or not) are basically just less severe and apply to smaller areas only. If the material is allowed to be used to cover the entire wall or ceiling, it is also allowed to be used to cover 10 percent of it. Although a Class C flame spread index may be lower than the flame spread index required for a particular building or facility, this quantity of combustible material will not significantly increase the fuel load.

804.2 Foam plastic interior trim. Foam plastic used as interior trim shall comply with Sections 804.2.1 through 804.2.4.

C As noted in Section 803.8.3, foam plastic is allowed as interior trim under certain conditions. This section introduces Sections 804.2.1 through 804.2.4, which establish that some dense foam plastic materials may be used as interior trim if the thickness, width and area of coverage is specifically limited.

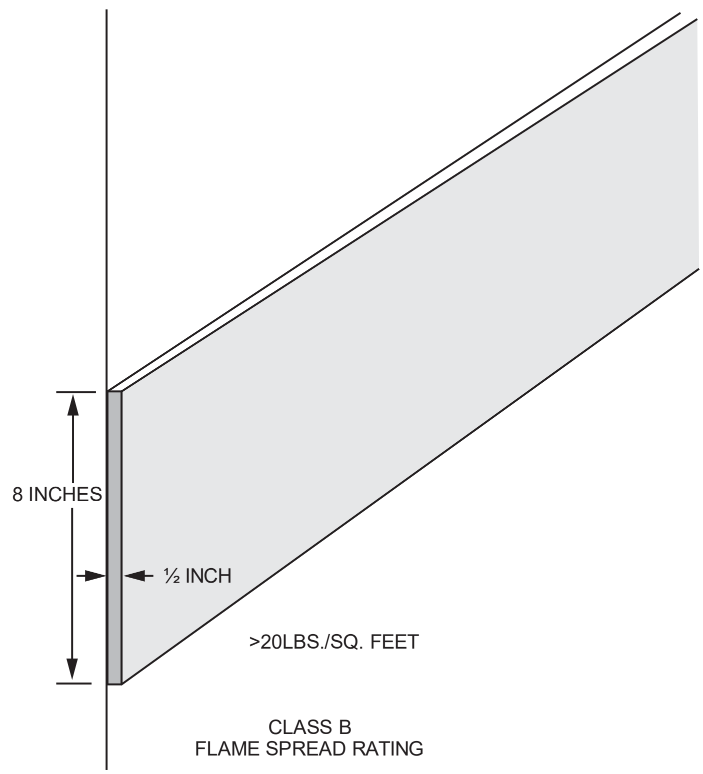
804.2.1 Density. The minimum density of the interior trim shall be 20 pounds per cubic foot (320 kg/m^3).

C This section establishes a minimum density of 20 pounds per cubic foot (pcf) (320 kg/m^3) for foam interior trim. A minimum instead of a maximum density is specified because the denser the foam plastic material, the less likely it is that it will generate misleading ASTM E84 test results due to having insufficient material for fire testing of the foam.

804.2.2 Thickness. The maximum thickness of the interior trim shall be $^1/_2$ inch (12.7 mm) and the maximum width shall be 8 inches (203 mm).

C Even though other trim materials are not limited in dimension, the maximum thickness and width of foam plastic trim is limited to 1/2 inch (12.7 mm) and 8 inches (203 mm), respectively. These dimensions were selected because they were typical of the maximums being produced at the time this provision was included in the code (see Commentary Figure 804.2.2).

Commentary Figure 804.2.2—Maximum Dimensions—Foam Plastic Trim

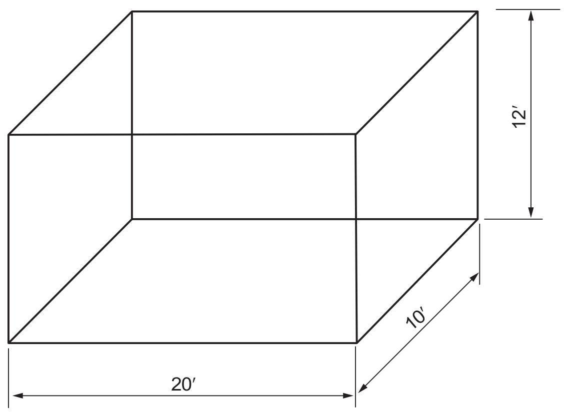


For SI: 1 foot = 304.8 mm, 1 pound per square foot = 4.882 kg/m^2.

804.2.3 Area limitation. The interior trim shall not constitute more than 10 percent of the specific wall or ceiling area to which it is attached.

C Trim cannot constitute more than 10 percent of the individual area of the wall and ceiling of a room. This limitation is simply a restatement of the general requirement for all combustible trim, which appears in Section 804.1 (see Commentary Figure 804.2.3).

Commentary Figure 804.2.3—Foam Plastic Decorative Trim Coverage Area



Wall area: 2 walls (20 × 12) + 2 walls (10 × 12) = 720 ft^2
10% of walls = 720 × 0.10 = 72 ft^2
Decorative trim coverage = 72 ft^2
Ceiling 20 × 10 = 200 ft^2
10% of ceiling = 200 × 0.10 = 20
Decorative trim coverage = 20 ft^2

For SI: 1 foot = 304.8 mm, 1 square foot = 0.0929 m^2.

804.2.4 Flame spread. The flame spread index shall not exceed 75 where tested in accordance with ASTM E84 or UL 723. The smoke-developed index shall not be limited.

Exception: Where the interior trim material has been tested as an interior finish in accordance with NFPA 286 and complies with the acceptance criteria in Section 803.1.1.1, it is not required to be tested for flame spread index in accordance with ASTM E84 or UL 723.

C This section specifically calls out a numerical flame spread index limitation of 75 for foam plastic used as trim, which is a Class B flame spread index. The value of 75 was selected to be consistent with the requirement for foam plastic insulation, even though other materials used as trim are permitted to have flame spread indexes of up to 200 in many locations. No thermal barrier is required for the use of foam plastic as trim and the smoke-developed index is not limited. It should be noted that IBC Section 2604, which regulates the use of foam plastic as trim for new construction, has essentially identical requirements to those found in this section. Neither this section nor IBC Section 2603.4.1.11 requires a thermal barrier for interior trim. The exception recognizes the stringency of testing in accordance with NFPA 286 as being equivalent to this section.

804.3 New interior floor finish. New interior floor finish and floor covering materials in new and existing buildings shall comply with Sections 804.3.1 through 804.3.3.2.

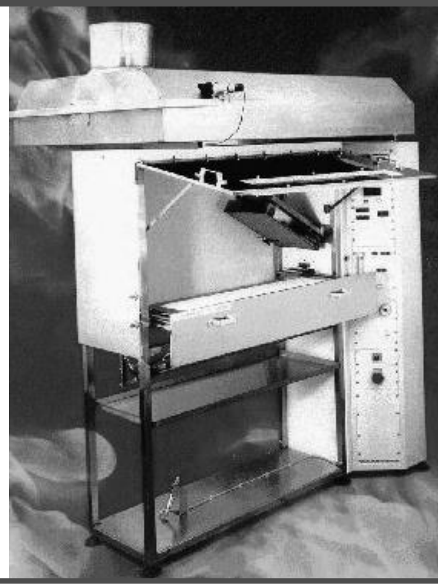
Exception: Floor finishes and coverings of a traditional type, such as wood, vinyl, linoleum or terrazzo, and resilient floor covering materials that are not composed of fibers.

C This section regulates the design and installation of floor finish and floor covering materials and is identical to IBC Section 804. Since Section 804.3 is located within a section that could apply to both existing and new buildings, it is specifically clarified that these requirements are only applicable to new flooring. Traditional floor coverings, such as wood, vinyl, terrazzo and other resilient floor covering material, must be exempt from this section since they generally contribute minimally to a fire. The focus is on textile floor coverings such as carpets.

804.3.1 Classification. Interior floor finish and floor covering materials required by Section 804.3.3.2 to be of Class I or II materials shall be classified in accordance with ASTM E648 or NFPA 253. The classification referred to herein corresponds to the classifications determined by ASTM E648 or NFPA 253 as follows: Class I, 0.45 watts/cm^2 or greater; Class II, 0.22 watts/cm^2 or greater.

C The use of a classification system eliminates the need to state the actual critical radiant flux value for a product to meet the identification requirements of Section 804.3. Over the years, a classification system has been found to be much easier for the industry to follow and still provides the building official with the information required to verify compliance. The test required to measure the combustibility of floor coverings can be either ASTM E648 or NFPA 253. This standard is a radiant floor panel test, which basically simulates materials subjected to heat from a fire above. The primary concern with flooring is related to the spread of a fire that has already been ignited within a space or room to a different room. Commentary Figure 804.3.1 shows the test apparatus. The critical heat flux indicates the threshold value above which flame spread occurs in the testing environment.

Commentary Figure 804.3.1—Radiant Floor Panel Test (NFPA 253)



804.3.2 Testing and identification. Interior floor finish and floor covering materials shall be tested by an *approved* agency in accordance with ASTM E648 or NFPA 253 and identified by a hang tag or other suitable method so as to identify the manufacturer or supplier and style, and shall indicate the interior floor finish or floor covering classification in accordance with Section 804.3.1. Carpet-type floor coverings shall be tested as proposed for use, including underlayment. Test reports confirming the information provided in the manufacturer's product identification shall be furnished to the *fire code official* upon request.

C The only method to ascertain that a floor meets the criteria of this section is to request a copy of the test report for the specific material being installed; therefore, it is critical that the carpeting be properly identified in order to verify that acceptable materials are being provided in the appropriate locations. The identification is to be provided on the material itself since a manufacturer's designation is required.

804.3.3 Interior floor finish requirements. New interior floor covering materials shall comply with Sections 804.3.3.1 and 804.3.3.2, and interior floor finish materials shall comply with Section 804.3.1.

C Sections 804.3.3.1 and 804.3.3.2 prescribe where the pill test for floor covering material is applicable and which occupancies and locations require a Class I or Class II classification in accordance with NFPA 253. Section 804.3.1 specifies the criteria for Classes I and II. More information about the specific test is discussed in Section 804.3.1. Generally, the focus is on more critical areas, such as exit passageways, interior exit stairways and exit access corridors.

804.3.3.1 Pill test. In all occupancies, new floor covering materials shall comply with the requirements of the DOC FF-1 "pill test" (CPSC 16 CFR Part 1630) or of ASTM D2859.

C DOC FF-1, also referred to as the "Methenamine Pill Test," was developed as a means of preventing the distribution of highly flammable soft floor coverings within the United States. The test essentially evaluates the performance of the floor covering when subject to a cigarette-type ignition by using a small methenamine tablet. All carpeting greater than 24 square feet (2.2 m^2) in area sold in the United States is required by federal law to pass this test procedure as a minimum.

804.3.3.2 Minimum critical radiant flux. In all occupancies, new interior floor finish and floor covering materials in enclosures for *stairways* and *ramps*, *exit passageways*, *corridors* and rooms or spaces not separated from *corridors* by full-height partitions extending from the floor to the underside of the ceiling shall withstand a minimum critical radiant flux. The minimum critical radiant flux shall be not less than Class I in Groups I-1, I-2 and I-3 and not less than Class II in Groups A, B, E, H, I-4, M, R-1, R-2 and S.

Exception: Where a building is equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1 or 903.3.1.2, Class II materials shall be permitted in any area where Class I materials are required and materials complying with DOC FF-1 "pill test" (CPSC 16 CFR Part 1630) or with ASTM D2859 shall be permitted in any area where Class II materials are required.

C This section prescribes the minimum requirements for interior floor finish materials that are used in interior exit stairways, exit passageways and exit access corridors. The criteria are based on the occupancy classification and the relationship of the space to the egress system. Similar to Table 803.3, the occupancy classification designation is meant to apply to the actual occupancy of the space and not necessarily the overall building classification.

Classifications I and II as used in this section are defined in Section 804.3.1 and are based on the results of the NFPA 253 test procedure.

Recognizing the ability of automatic sprinkler systems to control a fire and the minimal contribution of interior floor finishes to the early stages of fire growth, the exception allows the required interior floor finish ratings to be reduced where an automatic sprinkler system is provided throughout the building. The reference to Section 903.3.1.1 or 903.3.1.2 clarifies that the system is to be installed in accordance with NFPA 13 or 13R. In cases where Class II materials are required and an automatic sprinkler system is provided, the minimum requirement is that the material simply meet the DOC FF-1 test criteria.

804.4 Interior floor-wall base. Interior floor-wall base that is 6 inches (152 mm) or less in height shall be tested in accordance with ASTM E648 or NFPA 253 and shall be not less than Class II. Where a Class I floor finish is required, the floor-wall base shall be Class I. The classification referred to herein corresponds to the classifications determined by ASTM E648 or NFPA 253 as follows: Class I, 0.45 watt/cm^2 or greater; Class II, 0.22 watts/cm^2 or greater.

Exception: Interior trim materials that comply with Section 804.1.

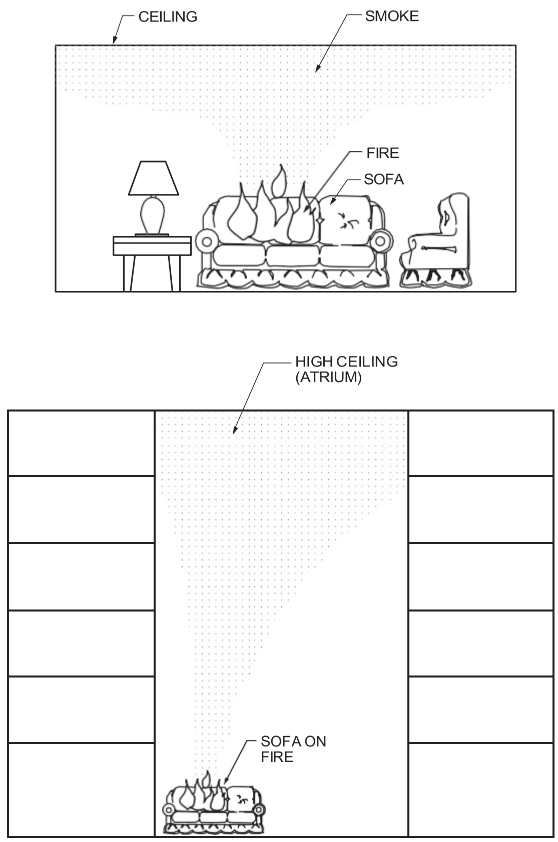
C In trimming out the interior of a building, rather than install separate baseboards or materials, in many cases the floor covering material is simply seamlessly turned up for a few inches or used at the intersection of the floor and the wall, thus becoming the floor-wall base trim. Previously, these materials could have been considered as interior trim in accordance with Section 804.1 and would have been required to be tested in accordance with ASTM E84 even though the floor covering may be required to be tested in accordance with ASTM E648 or NFPA 253. Based on the small amount of material used, it is very difficult to test these materials in a reliable manner, upside down in the ASTM E84 test method. This section addresses the issue of testing and regulation of wall base interior floor finish trim materials and eliminates that difficulty.

Because of their location at the floor line, wall base materials are not likely to be involved in a fire until the floor covering is also involved, usually at room flashover. Thus, it is reasonable that wall base materials meet the same criteria as floor coverings. This is true since in some applications, for sanitary reasons, the floor covering is seamlessly turned up on the wall to form a wall base. Thus, this section specifies that the wall base be tested in accordance with ASTM E648 or NFPA 253 as required for floor coverings and has requirements for its use. This section also limits the height of the wall base such that its application is controlled in a similar manner as the 10 percent limitation for interior trim. The exception recognizes that some materials used as interior finish trim that meet the flammability requirements of Section 804.1 can be used in this specific application without the need for additional testing.

SECTION 805—UPHOLSTERED FURNITURE AND MATTRESSES IN NEW AND EXISTING BUILDINGS

C Furnishings and contents are subjects that codes have not addressed very strongly in the past. Ultimately, the fire hazard potential within a building depends heavily on what is in the building and where it is placed. First, the burning characteristics of the materials will vary, and the location of the material will change the characteristics of a fire. For instance, a couch within a small compartment will create a much different fire event than the same couch burning in a large open atrium (see Commentary Figure 805.1). The compartment may be limited by the amount of oxygen available, while the atrium fire will be limited only by the amount of combustibles to burn because oxygen will be plentiful; therefore, the compartment is likely to reach flashover while the atrium will not. Generally, upholstered furniture and mattresses are the largest fire hazards in most residential buildings because mattresses or upholstered furniture are the only products typically present where people live that can cause room flashover on their own.

Commentary Figure 805.1—Effect of Location of Fire



805.1 Group I-1, Condition 2. The requirements in Sections 805.1.1 through 805.1.2 shall apply to facilities in Group I-1, Condition 2.

C This section introduces Sections 805.1.1 through 805.1.2.3, which contain requirements for controlling the hazards associated with upholstered furniture and mattresses in new and existing Group I-1, Condition 2 occupancies. Group I-1, Condition 2 occupancies contain, in a supervised setting, more than 16 persons on a 24-hour basis because of age, mental disability or other reasons. It should be noted that Group I-1, Condition 2 occupancies have occupants who need limited assistance both verbally and physically to evacuate whereas occupants of Group I-1, Condition 1 do not need such assistance. The provisions throughout the I-Codes differentiate between these two types of occupancy classifications. In general, these occupants are considered more vulnerable than the general population and have had a history of starting fires in beds or upholstered furniture. There is also more of a concern than with a Group I-2 occupancy over occupants having the ability to purposely start a fire. This is less likely in an assisted living setting but more likely in a halfway house setting; therefore, limitations on combustibility of upholstered furniture and mattresses are also required. Reducing ignitability and combustibility will significantly reduce the level of fire hazard. The requirements are slightly different than for Group I-2 occupancies in that different tests and performance criteria are required because of the nature of the hazards.

805.1.1 Upholstered furniture. Newly introduced upholstered furniture shall meet the requirements of Sections 805.1.1.1 through 805.1.1.3.

C This section informs the user that two aspects of upholstered furniture are regulated: the ignitability by cigarettes and the maximum allowable heat release in accordance with ASTM E1537.

805.1.1.1 Ignition by cigarettes. Newly introduced upholstered furniture shall be shown to resist ignition by cigarettes as determined by tests conducted in accordance with one of the following:

1. Mocked-up composites of the upholstered furniture shall have a char length not exceeding 1.5 inches (38 mm) when tested in accordance with NFPA 261.
2. The components of the upholstered furniture shall meet the requirements for Class I when tested in accordance with NFPA 260.

C This section focuses on the ignitability of furniture when exposed to lighted cigarettes and is consistent with Section 805.2.1.1 for Group I-2 occupancies and Section 805.3.1.1 for Group I-3 detention and correctional facilities. It offers an alternative test (NFPA 261) for approval of cigarette ignition resistance of newly introduced upholstered furniture in Group

I-1 occupancies (board and care facilities). The same test method is already permitted for use in Group I-2 and I-3 occupancies. The difference between NFPA 260 and NFPA 261 is that NFPA 260 uses an overall classification system that looks at the cigarette ignition behavior of the individual components that may be found in upholstered furniture while NFPA 261 specifically looks at ignitability of a mockup of upholstered furniture. It presents a method for study of the ignitability of the furniture mockup and a technique to measure the char length, and limits the maximum char length to 1.5 inches (38.1 mm). In fact, results from NFPA 261 are more likely to be predictive of real fire behavior.

The previously existing exception to this section for upholstered furniture in rooms or spaces protected by an approved NFPA 13 automatic sprinkler system has been deleted because: a. Sprinklers have no effect on controlling smoldering ignition (ignition by cigarettes), since they require an increase in room temperature to act and there will be no increase in room temperature until well after the upholstered furniture which fails the cigarette test has erupted into flames; and b. Newly introduced upholstered furniture is very likely to meet smoldering ignition requirements. The trade association for manufacturers of residential upholstered furniture, the Upholstered Furniture Action Council (UFAC), its sister organization, the American Home Furnishings Alliance (AHFA) and the trade association for manufacturers of institutional and contract upholstered furniture, the Business and Institutional Furniture Manufacturers Association (BIFMA), require that their members comply with the smoldering resistance test. UFAC requires NFPA 260 (equivalent to ASTM E1353 and the UFAC test) and BIFMA requires NFPA 261 (equivalent to ASTM E1352). Note that this section does not affect existing upholstered furniture.

805.1.1.2 Heat release rate. Newly introduced upholstered furniture shall have limited rates of heat release when tested in accordance with ASTM E1537 or California Technical Bulletin 133, as follows:

1. The peak rate of heat release for the single upholstered furniture item shall not exceed 80 kW.

   Exception: Upholstered furniture in rooms or spaces protected by an *approved automatic sprinkler system* installed in accordance with Section 903.3.1.1.

2. The total heat released by the single upholstered furniture item during the first 10 minutes of the test shall not exceed 25 megajoules (MJ).

   Exception: Upholstered furniture in rooms or spaces protected by an *approved automatic sprinkler system* installed in accordance with Section 903.3.1.1.

C The two tests specified in this section measure the overall combustibility of furniture. Where a building is sprinklered in accordance with NFPA 13, this section does not apply. Upholstered furniture must be tested to either ASTM E1537 or California Technical Bulletin 133 (which are basically the same test, but ASTM E1537 has no pass/fail criteria). This test uses a full-scale calorimeter, with the furniture item either in a standard room or under a hood. A full-scale calorimeter allows a representative piece of furniture to be burned and the products of combustion to be collected and analyzed in the exhaust duct by measuring gases (principally oxygen) in order to measure heat release. The test also measures weight loss during the test [see Commentary Figure 805.1.1.2(1) for a representation of the test]. The acceptance criteria set by the code are as follows:

- Peak heat release is limited to 80 kW.
- Total energy (or heat) release within the first 10 minutes cannot exceed 25 mega joules (MJ).

Limitations are placed on the maximum intensity and fire effluents produced by restricting the peak heat release rate and the amount of combustibles actually burned. The total energy release of 25 MJ could be translated to a steady fire as follows, where:

X = Steady fire heat release rate expressed in kW

X × 10 minutes = 25 MJ (10 minutes = 600 sec and 25 MJ = 25,000 kJ)

X × 600 sec = 25,000 kJ

X = 25,000 kJ/600 sec = 41.66 kJ/sec

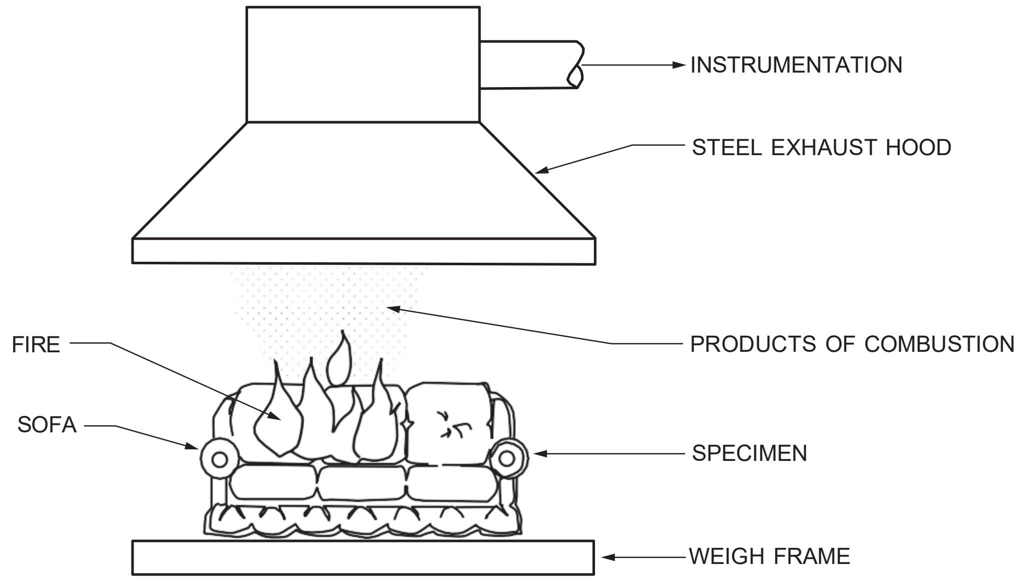
X = 41.66 kJ/sec = 41.66 kW ~ 42 kW heat release rate

A steady fire of 42 kW (133 kJ/sec) for 10 minutes will result in a total energy release of 25 MJ.

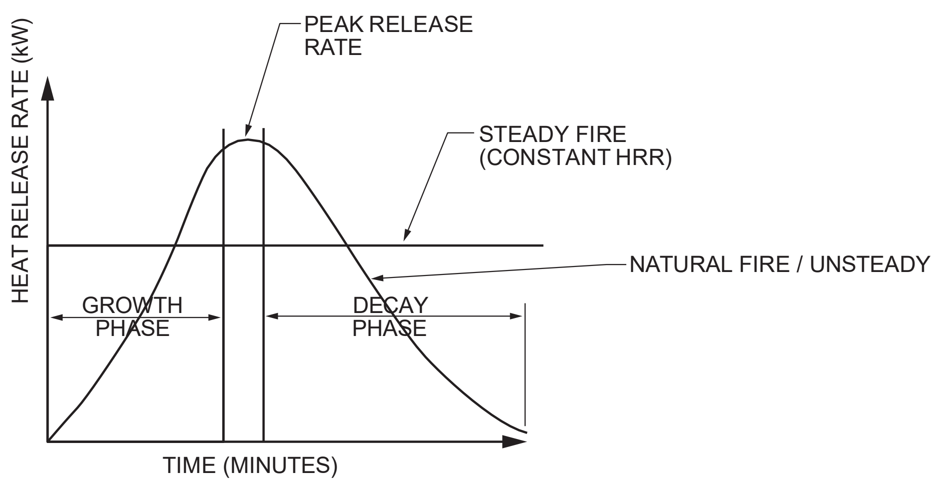
Because fires in more realistic conditions do not burn steadily and vary in their characteristics, the criterion is given in the form of a peak heat release rate and total energy release. To provide a better understanding, if the fire were burning at the maximum peak heat release rate of 80 kW for the first 10 minutes, the total energy output would be 75 MJ [80 kW (kJ/sec) H 10 minutes (600 sec) = 48,000 kJ = 48 MJ], which is well over the criterion of 25 MJ. A fire burning at a steady rate from start to finish is not realistic because fires must go through an initial growth stage before a peak heat release rate will be reached, followed by a decay phase; therefore, because a realistic fire will not burn at the peak heat release rate from the start of the fire, it is possible for a piece of furniture to have a peak heat release rate of 80 kW and still stay within the 25 MJ limitation. Commentary Figure 805.1.1.2(2) demonstrates the difference between a steady fire and a more realistic unsteady fire. As stated in the exceptions, the heat release rates will not apply to buildings sprinklered in accordance with NFPA 13. Sections 805.2.1.2 and 805.3.1.2 address the heat release rate limitations for upholstered furniture in the same manner. It should be noted that Section 805.3.1.2 does not reference California Technical Bulletin 133.

See also the commentary for mattresses in Section 805.1.2.

Commentary Figure 805.1.1.2(1)—Typical Fire Test Setup



Commentary Figure 805.1.1.2(2)—Steady Versus Unsteady Fires



805.1.1.3 Identification. Upholstered furniture shall bear the label of an *approved* agency, confirming compliance with the requirements of Sections 805.1.1.1 and 805.1.1.2.

C In order to achieve verifiable compliance, labeling by an approved agency is required. Otherwise, this information would be extremely difficult to verify in the field. See the commentary for the definition of "Labeled" in Section 202.

805.1.2 Mattresses. Newly introduced mattresses shall meet the requirements of Sections 805.1.2.1 through 805.1.2.3.

C Sections 805.1.2.1 and 805.1.2.2 deal with the combustibility of mattresses. Section 805.1.2.1 focuses on initial ignition and the ability of a mattress to sustain a fire; Section 805.1.2.2 is focused primarily on the burning characteristics of mattresses.

805.1.2.1 Ignition by cigarettes. Newly introduced mattresses shall be shown to resist ignition by cigarettes as determined by tests conducted in accordance with DOC 16 CFR Part 1632 and shall have a char length not exceeding 2 inches (51 mm).

C This section sets a maximum char length of 2 inches (51 mm) when the mattress is tested under DOC 16 CFR, Part 1632. This test is a mandatory regulation for all mattresses sold in the United States. More specifically, it is part of the regulations governed by the Consumer Products Safety Commission (CPSC) under the Department of Commerce (DOC). Sections 805.2.2.1 and 805.3.2.1 have the same reference and requirements for Group I-2 and I-3 occupancies. Thus, mattresses that fail to meet the 16 CFR 1632 test will be those sold either before the CPSC regulation went into effect (in 1972) or outside of the United States.

805.1.2.2 Heat release rate. Newly introduced mattresses shall have limited rates of heat release when tested in accordance with ASTM E1590 or California Technical Bulletin 129, as follows:

1. The peak rate of heat release for the single mattress shall not exceed 100 kW.

   Exception: Mattresses in rooms or spaces protected by an *approved automatic sprinkler system* installed in accordance with Section 903.3.1.1.

2. The total heat released by the single mattress during the first 10 minutes of the test shall not exceed 25 MJ.

   Exception: Mattresses in rooms or spaces protected by an *approved automatic sprinkler system* installed in accordance with Section 903.3.1.1.

C As noted, an occupant smoking in bed and initiating a mattress fire is a major fire hazard in Group I-1, Condition 2 occupancies. This section, like Section 805.1.1.2, limits combustibility by restricting the peak heat release rate and on the total energy output in the first 10 minutes of burning. These limitations vary slightly from those for upholstered furniture. Each mattress is allowed a maximum heat release rate of 100 kW, whereas each upholstered furniture item is limited to 80 kW. The total energy (or heat) release limitation is the same as for upholstered furniture items, which is 25 MJ in the first 10 minutes. As with upholstered furniture, these restrictions are not applicable in buildings sprinklered in accordance with NFPA 13. The tests that determine the peak heat release rate and total heat release are specific to mattresses. These tests are detailed in ASTM E1590 and California Technical Bulletin 129 (which are the same test, except that ASTM E1590 does not have pass/fail criteria). These tests, like that referenced in Section 805.1.1.2, make use of a full-scale calorimeter to measure the products of combustion. There are two differences between ASTM E1537 and ASTM E1590: the object being tested (see commentary, Section 805.1.1.2) and the ignition source. Both tests use a gas burner, but they are different in geometry, gas flow rate, duration of gas flow and position of flame application. Section 805.2.2.2 contains the same restrictions and testing requirements on combustibility for mattresses in Group I-2 occupancies, Section 805.3.2.2 for Group I-3 detention and correctional facilities and Section 805.4.2.2 for Group R-2 dormitories.

805.1.2.3 Identification. Mattresses shall bear the label of an *approved* agency, confirming compliance with the requirements of Sections 805.2.2.1 and 805.2.2.2.

C This section provides the fire code official with a valuable tool in evaluating and approving mattresses in Group I-2 occupancies. In order to achieve verifiable compliance, labeling by an approved agency is required. Otherwise, this information would be extremely difficult to verify in the field. See the commentary for the definition of "Labeled" in Section 202.

805.2 Group I-2 and Group B ambulatory care facilities. The requirements in Sections 805.2.1 through 805.2.2 shall apply to Group I-2 occupancies and Group B ambulatory care facilities.

C This section introduces Sections 805.2.1 through 805.2.2.3, which contain requirements for controlling the hazards associated with upholstered furniture and mattresses in new and existing Group I-2 occupancies and Group B ambulatory care facilities. Occupants of nursing homes (Group I-2, Condition 1), hospitals (Group I-2, Condition 2) and ambulatory care facilities (Group B) are considered more vulnerable than the general population. Many of the patients of these occupancies are confined because of respirators, IVs and other medical equipment, and may not be capable of self-preservation. Hospital employees may have to make several trips into the fire area to assist in evacuating patients; therefore, it is imperative that every effort be made to preserve the integrity of the corridors and minimize the fuel loading caused by furnishings and contents, such as upholstered furniture and mattresses.

Historically, fires have been ignited through the use of cigarettes in bed or falling asleep while smoking in a chair. In addition to these hazards, such occupancies usually have additional medical oxygen sources within their rooms. This section, therefore, states several ignitability and combustibility limitations for upholstered furniture and mattresses that might be introduced into such occupancies.

805.2.1 Upholstered furniture. Newly introduced upholstered furniture shall meet the requirements of Sections 805.2.1.1 through 805.2.1.3.

C This section informs the user that two aspects of upholstered furniture are regulated: the ignitability by cigarettes and the maximum allowable heat release in accordance with ASTM E1537.

805.2.1.1 Ignition by cigarettes. Newly introduced upholstered furniture shall be shown to resist ignition by cigarettes as determined by tests conducted in accordance with one of the following: (a) mocked-up composites of the upholstered furniture shall have a char length not exceeding 1.5 inches (38 mm) when tested in accordance with NFPA 261 or (b) the components of the upholstered furniture shall meet the requirements for Class I when tested in accordance with NFPA 260.

Exception: Upholstered furniture belonging to the patients in sleeping rooms of Group I-2, Condition 1 occupancies, provided that a smoke detector is installed in such rooms. Battery-powered, single-station smoke alarms shall be allowed.

C The exception allows nursing home patients to bring their furniture with them into their rooms without having to comply with the test requirements, provided that the basic protection of a system smoke detector (in buildings equipped with a fire alarm system) or a battery-operated smoke alarm is present in the room (see commentary, Section 805.1.1.1). Note that all new Group I-2, Condition 1 occupancies are required to be equipped throughout with an automatic sprinkler system. Additionally, Section 1105.9 requires an automatic sprinkler system in existing Group I-2 occupancies. This requirement is retroactive, which means it applies regardless of whether work is being done to the building.

805.2.1.2 Heat release rate. Newly introduced upholstered furniture shall have limited rates of heat release when tested in accordance with ASTM E1537 or California Technical Bulletin 133, as follows:

1. The peak rate of heat release for the single upholstered furniture item shall not exceed 80 kW.

   Exception: Upholstered furniture in rooms or spaces protected by an *approved automatic sprinkler system* installed in accordance with Section 903.3.1.1.

2. The total heat released by the single upholstered furniture item during the first 10 minutes of the test shall not exceed 25 MJ.

   Exception: Upholstered furniture in rooms or spaces protected by an *approved automatic sprinkler system* installed in accordance with Section 903.3.1.1.

C This section is identical to Section 805.1.1.2 and focuses on the potential heat release rate of upholstered furniture if a fire occurs (see commentary, Section 805.1.1.2).

805.2.1.3 Identification. Upholstered furniture shall bear the label of an *approved* agency, confirming compliance with the requirements of Sections 805.2.1.1 and 805.2.1.2.

C In order to achieve verifiable compliance, labeling by an approved agency is required. Otherwise, this information would be extremely difficult to verify in the field. See the commentary to the definition of "Labeled" in Section 202.

805.2.2 Mattresses. Newly introduced mattresses shall meet the requirements of Sections 805.2.2.1 through 805.2.2.3.

C Sections 805.2.2.1 and 805.2.2.2 deal with the combustibility of mattresses. Section 805.2.2.1 focuses on initial ignition and the ability of a mattress to sustain a fire; Section 805.2.2.2 is focused primarily on the burning characteristics of mattresses.

805.2.2.1 Ignition by cigarettes. Newly introduced mattresses shall be shown to resist ignition by cigarettes as determined by tests conducted in accordance with DOC 16 CFR Part 1632 and shall have a char length not exceeding 2 inches (51 mm).

C This section is the same as Sections 805.1.2.1 and 805.3.2.1 and requires compliance with DOC 16 CFR Part 1632 (see commentary, Section 805.1.2.1).

805.2.2.2 Heat release rate. Newly introduced mattresses shall have limited rates of heat release when tested in accordance with ASTM E1590 or California Technical Bulletin 129, as follows:

1. The peak rate of heat release for the single mattress shall not exceed 100 kW.

   Exception: Mattresses in rooms or spaces protected by an *approved automatic sprinkler system* installed in accordance with Section 903.3.1.1.

2. The total heat released by the single mattress during the first 10 minutes of the test shall not exceed 25 MJ.

   Exception: Mattresses in rooms or spaces protected by an *approved automatic sprinkler system* installed in accordance with Section 903.3.1.1.

C This section is the same as Section 805.1.2.2 in limiting the maximum heat release rate and the total heat release (see commentary, Section 805.1.2.2).

805.2.2.3 Identification. Mattresses shall bear the label of an *approved* agency, confirming compliance with the requirements of Sections 805.2.2.1 and 805.2.2.2.

C In order to achieve verifiable compliance, labeling by an approved agency is required. Otherwise, this information would be extremely difficult to verify in the field. See the commentary to the definition of "Labeled" in Section 202.

805.3 Group I-3, detention and correction facilities. The requirements in Sections 805.3.1 through 805.3.2 shall apply to detention and correction facilities classified in Group I-3.

C This section introduces Sections 805.3.1 through 805.3.2.3, which contain requirements for controlling the hazards associated with upholstered furniture and mattresses in new and existing Group I-3 detention and correctional occupancies. These facilities have a higher likelihood of incendiary activities, and because of the restrictions on movement, the occupants are placed in a more vulnerable position than the general public; therefore, combustibility limitations are placed on furniture and mattresses.

805.3.1 Upholstered furniture. Newly introduced upholstered furniture shall meet the requirements of Sections 805.3.1.1 through 805.3.1.3.

C This section informs the user that two aspects of upholstered furniture are regulated: the ignitability by cigarettes and the maximum allowable heat release in accordance with ASTM E1537.

805.3.1.1 Ignition by cigarettes. Newly introduced upholstered furniture shall be shown to resist ignition by cigarettes as determined by tests conducted in accordance with one of the following:

1. Mocked-up composites of the upholstered furniture shall have a char length not exceeding 1.5 inches (38 mm) when tested in accordance with NFPA 261.

2. The components of the upholstered furniture shall meet the requirements for Class I when tested in accordance with NFPA 260.

C These requirements are the same as those found in Sections 805.1.1.1 and 805.2.1.1 (see commentary, Section 805.1.1.1).

805.3.1.2 Heat release rate. Newly introduced upholstered furniture shall have limited rates of heat release when tested in accordance with ASTM E1537, as follows:

1. The peak rate of heat release for the single upholstered furniture item shall not exceed 80 kW.
2. The total heat released by the single upholstered furniture item during the first 10 minutes of the test shall not exceed 25 MJ.

C See the commentary to Section 805.1.1.2.

805.3.1.3 Identification. Upholstered furniture shall bear the label of an *approved* agency, confirming compliance with the requirements of Sections 805.3.1.1 and 805.3.1.2.

C In order to achieve verifiable compliance, labeling by an approved agency is required. Otherwise, this information would be extremely difficult to verify in the field. See the commentary to the definition of "Labeled" in Section 202.

805.3.2 Mattresses. Newly introduced mattresses shall meet the requirements of Sections 805.3.2.1 through 805.3.2.3.

C Sections 805.3.2.1 and 805.3.2.2 deal with the combustibility of mattresses. Section 805.3.2.1 focuses on initial ignition and the ability of a mattress to sustain a fire; Section 805.3.2.2 focuses primarily on the burning characteristics of mattresses.

805.3.2.1 Ignition by cigarettes. Newly introduced mattresses shall be shown to resist ignition by cigarettes as determined by tests conducted in accordance with DOC 16 CFR Part 1632 and shall have a char length not exceeding 2 inches (51 mm).

C This section is the same as Sections 805.1.2.1 and 805.3.2.1 and requires compliance with DOC 16 CFR Part 1632 (see commentary, Section 805.1.2.1).

805.3.2.2 Fire performance tests. Newly introduced mattresses shall be tested in accordance with Section 805.3.2.2.1 or 805.3.2.2.2.

C This section provides two options for fire tests: ASTM E1590, which is a cone calorimeter test, and Annex A3 of ASTM F1085, which provides a simple burn test to determine how much mass the mattress will lose with this exposure.

805.3.2.2.1 Heat release rate. Newly introduced mattresses shall have limited rates of heat release when tested in accordance with ASTM E1590 or California Technical Bulletin 129, as follows:

1. The peak rate of heat release for the single mattress shall not exceed 100 kW.
2. The total heat released by the single mattress during the first 10 minutes of the test shall not exceed 25 MJ.

C This section is the same as Section 805.1.2.2 in limiting the maximum heat release rate and the total energy heat release (see commentary, Section 805.1.2.2).

805.3.2.2.2 Mass loss test. Newly introduced mattresses shall have a mass loss not exceeding 15 percent of the initial mass of the mattress where tested in accordance with the test in Annex A3 of ASTM F1085.

C The test in Annex A3 of ASTM F1085 was developed originally for use in detention and correctional occupancies and is a very severe test that is a reasonable (and less expensive) alternative to ASTM E1590. This test is very simple, can be conducted at any facility and does not require the use of an instrumented fire test lab. The test can be described in a few words: it involves rolling up a mattress, placing it at an angle (for example by holding it with a brick), introducing newspaper into the volume surrounding the rolled up mattress and igniting the newspaper with a match.

If mattress materials melt away from the flame with flaming drips they may "pass" the ASTM E1590 test; however, melting would result in a failure of the ASTM F1085 Annex A test. In this test, the material that flames on the floor will keep burning the mattress itself.

Commentary Figure 805.3.2.2.2 provides a table that shows the results using the ASTM F1085 Annex A3 test for a number of mattresses in two studies (one in 1980 and one in 2003) and it also shows whether the mattresses meet the ASTM E1590 requirements in the code. It is clear from the table that mattresses will either burn up almost completely or lose very little mass. The ASTM F1085 test will not pass mattresses that fail ASTM E1590.

The test from Annex A3 of ASTM F1085 is also described in Section 10.2 of ASTM F1870 *(Standard Guide for Selection of Fire Test Methods for the Assessment of Upholstered Furnishings in Detention and Correctional Facilities)* as a test method "Designed for Detention and Correction Facilities."

Commentary Figure 805.3.2.2.2—Annex A3 ASTM F1085 Test Results

MATTRESS	ASTM F1085 MASS LOSS %	ASTM E1590 PER SECTION 805.3.2.2.1 PASS OR FAIL
1 (2003)	1.22	Pass
2 (2003)	9.47	Pass
3 (2003)	3.30	Pass
4 (2003)	100	Fail
5 (2003)	100	Fail
1 (1980)	100	Fail
2 (1980)	100	Fail
3 (1980)	98.5	Fail
4 (1980)	91.1	Fail
5 (1980)	91	Fail
6 (1980)	83.1	Fail
7 (1980)	44.7	Fail
8 (1980)	3.0	Pass

805.3.2.3 Identification. Mattresses shall bear the label of an *approved* agency, confirming compliance with the requirements of Sections 805.3.2.1 and 805.3.2.2.

C In order to achieve verifiable compliance, labeling by an approved agency is required. Otherwise, this information would be extremely difficult to verify in the field. See the commentary to the definition of "Labeled" in Section 202.

805.4 Group R-2 college and university dormitories. The requirements of Sections 805.4.1 through 805.4.2.3 shall apply to college and university dormitories classified in Group R-2, including decks, porches and balconies.

C This section introduces Sections 805.4.1 through 805.4.2.3, which contain requirements for controlling the hazards associated with upholstered furniture and mattresses in new and existing Group R-2 college and university dormitories. Fire represents a significant risk to life and property in dormitory occupancies, particularly at residential schools, colleges and universities. The large number of young people living in close proximity to one another creates the potential for a relatively small fire to have serious and possibly fatal consequences. According to NFPA fire loss data, there are an average of 1,425 fires each year in dormitories causing $6.3 million in direct property damage. For these reasons, upholstered furniture and mattresses in dormitories must comply with the same requirements for fire performance as those in Group I-1, I-2 and I-3 occupancies. The test methods and criteria are identical to those in Sections 805.1, 805.2 and 805.3. Section 805.3.2.2.2 provides an option for a mass loss test for mattresses.

Note that this section also includes upholstered furniture and mattresses on porches, decks and balconies, which, for this particular occupancy, can be frequently found in such locations. Since the decks, porches or balconies are likely not sprinklered, this would severely limit or eliminate the allowance of such items in these areas. Many jurisdictions have specifically banned furniture such as couches from these areas. These provisions are not as strict but will limit the potential for fires to spread to the structure.

Section 805.4 provides the fire code official and college and university campus housing authorities with the needed authority to limit the combustibility of student-owned furnishings. Those furnishings can include both upholstered furniture and mattresses, which are the high-fuel items in Group R-2 dormitory occupancies. Strictly enforced, these provisions, coupled with the limitations on the use of open flames in Group R-2 dormitories found in Section 308.4.1, should provide a much-improved level of fire safety in these occupancies.

805.4.1 Upholstered furniture. Newly introduced upholstered furniture shall meet the requirements of Sections 805.4.1.1 through 805.4.1.3.

C This section informs the user that two aspects of upholstered furniture are regulated: the ignitability by cigarettes and the maximum allowable heat release in accordance with recognized standards.

805.4.1.1 Ignition by cigarettes. Newly introduced upholstered furniture shall be shown to resist ignition by cigarettes as determined by tests conducted in accordance with one of the following:

1. Mocked-up composites of the upholstered furniture shall have a char length not exceeding $1^1/_2$ inches (38 mm) when tested in accordance with NFPA 261.

2. The components of the upholstered furniture shall meet the requirements for Class I when tested in accordance with NFPA 260.

C These requirements are the same as those found in Sections 805.1.1.1, 805.2.1.1 and 805.3.1.1 (see commentary, Section 805.1.1.1).

805.4.1.2 Heat release rate. Newly introduced upholstered furniture shall have limited rates of heat release when tested in accordance with ASTM E1537 or California Technical Bulletin 133, as follows:

1. The peak rate of heat release for the single upholstered furniture item shall not exceed 80 kW.

   Exception: Upholstered furniture in rooms or spaces protected by an *approved automatic sprinkler system* installed in accordance with Section 903.3.1.1.
2. The total heat released by the single upholstered furniture item during the first 10 minutes of the test shall not exceed 25 MJ.

   Exception: Upholstered furniture in rooms or spaces protected by an *approved automatic sprinkler system* installed in accordance with Section 903.3.1.1.

C This section is identical to Sections 805.1.1.2 and 805.2.1.2 and focuses on the potential heat release rate of upholstered furniture if a fire occurs (see commentary, Section 805.1.1.2).

805.4.1.3 Identification. Upholstered furniture shall bear the label of an *approved* agency, confirming compliance with the requirements of Sections 805.4.1.1 and 805.4.1.2.

C In order to achieve verifiable compliance, labeling by an approved agency is required. Otherwise, this information would be extremely difficult to verify in the field. See the commentary for the definition of "Labeled" in Section 202.

805.4.2 Mattresses. Newly introduced mattresses shall meet the requirements of Sections 805.4.2.1 through 805.4.2.3.

C Sections 805.4.2.1 and 805.4.2.2 deal with the combustibility of mattresses. Section 805.4.2.1 focuses on initial ignition and the ability of a mattress to sustain a fire; Section 805.4.2.2 focuses primarily on the burning characteristics of mattresses.

805.4.2.1 Ignition by cigarettes. Newly introduced mattresses shall be shown to resist ignition by cigarettes as determined by tests conducted in accordance with DOC 16 CFR Part 1632 and shall have a char length not exceeding 2 inches (51 mm).

C This section is the same as Sections 805.1.2.1, 805.2.2.1 and 805.3.2.1 and requires compliance with DOC 16 CFR Part 1632 (see commentary, Section 805.1.2.1).

805.4.2.2 Heat release rate. Newly introduced mattresses shall have limited rates of heat release when tested in accordance with ASTM E1590 or California Technical Bulletin 129, as follows:

1. The peak rate of heat release for the single mattress shall not exceed 100 kW.

   Exception: Mattresses in rooms or spaces protected by an *approved automatic sprinkler system* installed in accordance with Section 903.3.1.1.
2. The total heat released by the single mattress during the first 10 minutes of the test shall not exceed 25 MJ.

   Exception: Mattresses in rooms or spaces protected by an *approved automatic sprinkler system* installed in accordance with Section 903.3.1.1.

C This section is the same as Sections 805.1.2.2, 805.2.2.2 and 805.3.2.2.1 in limiting the maximum heat release rate and the total heat release (see commentary, Section 805.1.2.2).

805.4.2.3 Identification. Mattresses shall bear the label of an *approved* agency, confirming compliance with the requirements of Sections 805.4.2.1 and 805.4.2.2.

C In order to achieve verifiable compliance, labeling by an approved agency is required. Otherwise, this information would be extremely difficult to verify in the field. See the commentary to the definition of "Labeled" in Section 202.

SECTION 806—NATURAL DECORATIVE VEGETATION IN NEW AND EXISTING BUILDINGS

806.1 Natural cut trees. Natural cut trees, where allowed by this section, shall have the trunk bottoms cut off not less than 0.5 inch (12.7 mm) above the original cut and shall be placed in a support device complying with Section 806.1.2.

C This section focuses on the fire hazards posed by vegetation within a building. The majority of this section deals specifically with fresh Christmas trees. Natural cut trees, such as Christmas trees, within buildings pose a significant fire threat to occupants if they are not properly cared for. Because of the symmetric way in which these trees are normally groomed, the large surface area of the foliage and the amount of airspace throughout the branches, fires have the potential to burn very efficiently and vigorously if the tree is dry. The National Fire Protection Association (NFPA) reports that there are an estimated annual average of 200 home structure fires that begin with Christmas trees. Based on data from 2011 through 2015, these fires caused an average of 6 civilian deaths, 16 civilian injuries, and $14.8 million in direct property damage per year. The NFPA analysis also shows that although the number of Christmas tree fires is low, these fires represent a higher level of hazard. On average, one of every 32 Christmas tree fires resulted in a fatality compared to an average of one death per 143 nonconfined home structure fires overall. Further, 69 percent of Christmas tree fires spread beyond the room of origin. The fires that spread beyond the room of origin caused all of the associated fatalities. The percentage of trees involved in structure fires represents an extremely small portion of the total number of natural Christmas trees sold in the United States each year, which is estimated at 20 to 30 million. The moisture content of each tree can play a dominant role in determining the fire hazard it represents. Properly maintaining a cut Christmas tree is important to retaining a high moisture content in the needles of the tree to limit accidental ignition and prevent rapid flame spread. A tree that has dry needles can readily ignite with a flaming source and generate heat release rates that are capable of causing flashover in residential scale rooms. The above statistics addressed home structure fires and natural Christmas trees, which are required by the IFC to be kept moist and fresh at all times.

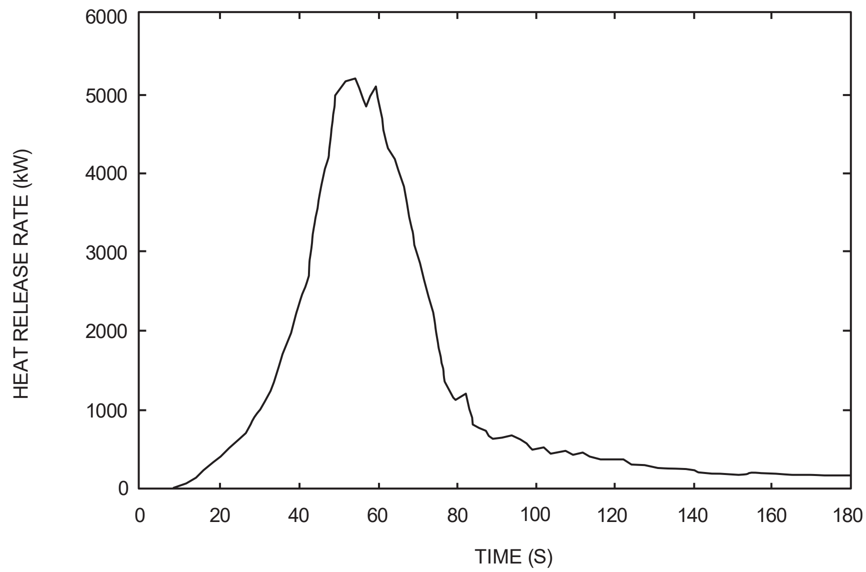
In tests performed at NIST in 1999 to better understand the severity of Christmas tree fires, eight Scotch pine Christmas trees were placed in a room at 73°F (23°C) at approximately 50 percent relative humidity for approximately three weeks. Seven of the eight trees were given no additional moisture; the eighth tree was watered according to local rules for trees within business occupancies. This required the tree to be cut in a certain manner and placed in a stand with at least a 2-gallon (7.6 L) capacity. The seven dry trees were ignited and burned intensely; however, the tree that had been watered continuously throughout the three weeks could not be ignited. The peak heat releases from the dry trees ranged from approximately 1,600 kW to 5,000 kW within about 50 to 80 seconds of ignition [see Commentary Figure 806.1(1) for a summary of results and Commentary Figure 806.1(2) for a graphical representation of Test No. 3]. These test results demonstrate that the proper care for trees (i.e., making sure that the tree remains humid or wet) makes a significant difference on the fire hazard presented. This section requires at least 1/2 inch (12.7 mm) of tree trunk to be removed above the original cut to optimize the tree's ability to absorb water to maintain a minimum level of moisture. This section also requires a support device (tree stand or equivalent) that meets the criteria of Section 806.1.2. Such devices are intended to bring the correct amount of moisture into contact with the tree. The specifics of this section cannot realistically be monitored in all buildings within a jurisdiction. The focus must be on occupancies such as assembly or mercantile. All occupancies benefit, however, because there is now a method that a fire department could use to educate the general public on the treatment of Christmas trees. This tool can be helpful in fire prevention within a jurisdiction.

Commentary Figure 806.1(1)—Summary of Tree Parameters

TEST NO.	WEIGHT (kg)		HEIGHT (m)	WIDTH[a] (m)	MOISTURE CONTENT (%)
	Before Test	After Test			
1	17.2	6.8	2.6	1.7	30
2	15.9	8.2	2.7	1.3	27
3	20.0	6.8	2.3	1.7	30
4	9.5	5.0	2.5	1.2	30
5	19.1	8.6	2.5	1.7	28
6	12.7	7.7	2.5	1.1	32
7	18.6	7.7	3.1	1.5	25
8	28.1	28.1	2.7	1.4	36

(From NIST Report FR 4010)
a. The width is measured at the widest point of the tree.

Commentary Figure 806.1(2)—Graph of Heat Release Rate Versus Time for Test Number 3



(From NIST Report FR 4010)

806.1.1 Restricted occupancies. Natural cut trees shall be prohibited within ambulatory care facilities and Group A, E, I-1, I-2, I-3, I-4, M, R-1, R-2 and R-4 occupancies.

Exceptions:

1. Trees located in areas protected by an *approved automatic sprinkler system* installed in accordance with Section 903.3.1.1 or 903.3.1.2 shall not be prohibited in Groups A, E, M, R-1 and R-2.
2. Trees shall be allowed within *dwelling units* in Group R-2 occupancies.

C Although trees that have been properly watered and handled are less hazardous, there is still a concern that they should not be located in certain occupancies and specific uses. The occupancies listed in this section are those where the occupant load is high, occupants are vulnerable or the potential hazards that exist if the tree is not properly handled are too great. These occupancies and specific uses include ambulatory care facilities and Group A, E, I-1, I-2, I-3, I-4, M, R-1 and R-2 occupancies. There are exceptions for Group A, E, M, R-1 and R-2 occupancies when the trees are located in areas that are sprinklered in accordance with NFPA 13 or 13R, as applicable, and for trees within individual dwelling units in Group R-2. Essentially, the only occupancies that would be completely prohibited from having cut trees are institutional occupancies because of the vulnerability of the occupants and, in some cases, concerns for incendiary tendencies of some occupants.

806.1.2 Support devices. The support device that holds the tree in an upright position shall be of a type that is stable and that meets all of the following criteria:

1. The device shall hold the tree securely and be of adequate size to avoid tipping over of the tree.
2. The device shall be capable of containing a minimum two-day supply of water.
3. The water level, when full, shall cover the tree stem not less than 2 inches (51 mm). The water level shall be maintained above the fresh cut and checked not less than once daily.

C This section is intended to prevent the tree from tipping over, potentially into an ignition source, and ensure that the support device is designed and used to keep the water supply to the tree at a useful level. More specifically, Item 1 requires the device to prevent the tree from tipping; Item 2 states that the device must be capable of holding a two-day water supply; and Item 3 requires a minimum coverage of water to 2 inches (51 mm) above the bottom of the stem. The water level must be checked at least once daily to verify that an adequate supply remains. The restrictions in Section 806.1.1 are in place because the provisions in Section 806.1.2 are generally difficult for a fire department to monitor. Having a sprinkler system or generally prohibiting natural cut trees in higher risk occupancies provides a redundancy to deal with the potential hazards.

806.1.3 Dryness. The tree shall be removed from the building whenever the needles or leaves fall off readily when a tree branch is shaken or if the needles are brittle and break when bent between the thumb and index finger. The tree shall be checked daily for dryness.

C Lack of moisture is the primary problem with natural cut trees within buildings. This section describes a daily test to assist in evaluating whether the tree is considered too dry to remain in the building.

806.1.4 Fire-retardant treatments for natural cut trees. Where fire-retardant treatments are applied to natural cut trees, the fire-retardant treatment shall be tested by an *approved* agency and shall comply with both Test Method 1 and Test Method 2 of ASTM E3082.

C This section clarifies that the test requirement only becomes a requirement where fire retardant treatments are applied. This means that it does not introduce a requirement that any treatment be applied to natural cut trees. Note that the requirement applies to the treatment itself and does not apply to the tree. Therefore, it is the commercial treatment that needs to be tested and not the individual trees.

806.2 Obstruction of means of egress. The required width of any portion of a *means of egress* shall not be obstructed by decorative vegetation. Natural cut trees shall not be located within an *exit*, *corridor*, or a lobby or vestibule.

C Decorative vegetation is often placed in a location that does not normally accommodate combustibles. This section restricts locations so that the vegetation does not block the egress width and increase the fire hazard in such areas. For example, this would likely prohibit trees in main lobby areas of a movie theater.

806.3 Open flame. Candles and open flames shall not be used on or near decorative vegetation. Natural cut trees shall be kept a distance from heat vents and any open flame or heat-producing devices not less than the height of the tree.

C This section addresses the primary ignition hazards associated with decorative vegetation. More specifically, candles should never be placed on or near Christmas trees. Also, heat sources such as vents may pose an ignition hazard in addition to being a source of airflow that could dry out a tree, making it more susceptible to ignition.

806.4 Electrical fixtures and wiring. The use of unlisted electrical wiring and lighting on natural vegetation, including natural cut trees, shall be prohibited.

C Decorations are quite often used year after year on Christmas trees. Some of this decor consists of lights that are not specifically listed and can fail and become very hot, potentially posing an ignition source. Additionally, this wiring may arc and create an ignition source. In fact, it has been found that many Christmas tree fires have actually been started by decorative lights on the tree. Therefore, this section simply prohibits the use of unlisted wiring or lighting (such lights and ornaments are listed in accordance with UL 588).

SECTION 807—DECORATIVE MATERIALS AND ARTIFICIAL DECORATIVE VEGETATION IN NEW AND EXISTING BUILDINGS

807.1 General. The following requirements shall apply to all occupancies:

1. Furnishings or decorative materials of an explosive or highly flammable character shall not be used.
2. Fire-retardant coatings in existing buildings shall be maintained so as to retain the effectiveness of the treatment under service conditions encountered in actual use.
3. Furnishings or other objects shall not be placed to obstruct *exits*, access thereto, egress therefrom or visibility thereof.
4. The permissible amount of noncombustible decorative materials shall not be limited.

C The requirements in this section apply to decorative materials and artificial decorative vegetation in new and existing buildings. The bulk of the requirements in this section are applicable to all occupancy groups. Section 807.2 is applicable to the occupancy groups listed. In the specific case of Group I-3, combustible decorative materials are prohibited (See Section 807.5.4). Decorative materials must be noncombustible or meet the flame propagation performance criteria of Section 806.4 and NFPA 701. Note that Section 807.5 is occupancy based and addresses combustible decorative materials that may not comply with Section 807.3, such as artwork in classrooms. Instead, specific limitations on amounts and types of materials are provided. The occupancies addressed include Group A, E, I and R-2 dormitories.

This section addresses some basic hazards with certain finishes and materials associated with decorative materials. These are discussed below.

1. This is a general statement prohibiting furnishings and contents or decorative materials that pose an extreme fire potential. These materials tend to have a significant impact on fire size in a building.
2. In many cases, the IBC and this code allow use of certain combustible or flammable materials based on the application of a fire-retardant coating (see commentary, Section 803.4).
3. Similar to Section 806.3, this section prohibits any decorations or other objects from obstructing the means of egress, including visibility.
4. It is clarified that noncombustible decorative materials are not limited, as they pose no fire hazard. It is important that they comply with Item 3 in terms of obstruction of the means of egress.

807.2 Combustible decorative materials. In Groups A, B, E, I, M and R-1 and in dormitories in Group R-2, curtains, draperies, fabric hangings and other similar combustible decorative materials suspended from walls or ceilings shall comply with Section 807.3 and shall not exceed 10 percent of the specific wall or ceiling area to which such materials are attached.

Fixed or movable walls and partitions, paneling, wall pads and crash pads applied structurally or for decoration, acoustical correction, surface insulation or other purposes shall be considered to be *interior finish*, shall comply with Section 803 and shall not be considered *decorative materials* or furnishings.

Exceptions:

1. In auditoriums in Group A, the permissible amount of curtains, draperies, fabric hangings and similar combustible decorative material suspended from walls or ceilings shall not exceed 75 percent of the aggregate wall area where the building is equipped throughout with an *approved automatic sprinkler system* in accordance with Section 903.3.1.1, and where the material is installed in accordance with Section 803.15 of the *International Building Code.*
2. In Group R-2 dormitories, within *sleeping units* and *dwelling units*, the permissible amount of curtains, draperies, fabric hangings and similar decorative materials suspended from walls or ceilings shall not exceed 50 percent of the aggregate wall areas where the building is equipped throughout with an *approved automatic sprinkler system* installed in accordance with Section 903.3.1.
3. In Group B and M occupancies, the amount of combustible fabric partitions suspended from the ceiling and not supported by the floor shall comply with Section 807.3 and shall not be limited.
4. The 10 percent limit shall not apply to curtains, draperies, fabric hangings and similar combustible decorative materials used as window coverings.

C Meeting the flame propagation performance criteria of NFPA 701, which is required by Section 807.4, does not mean that materials will not burn, only that they are going to spread flame relatively slowly. These materials are, therefore, limited to a maximum of 10 percent of the total wall or ceiling area of the space under consideration. Unlike incidental trim, decorative materials are not necessarily distributed evenly throughout the room. Additionally, consideration of the long-term maintenance of the materials, including possible periodic retreatment, should be taken into account.

In any occupancy classification, when a movable wall, partition, paneling, wall pads or crash pads cover larger areas, they need to be dealt with as interior finishes instead of as decorative materials.

The first two exceptions relate to the 10 percent limitation in Section 807.3. Exception 1 is for Group A auditoriums that would allow 75 percent coverage of the walls and ceilings (instead of the limit of 10 percent) if the space is sprinklered in accordance with NFPA 13 and the material is applied in accordance with IBC Section 803.11.

Exception 2 allows an increased amount of decorative materials in Group R-2 dormitories. An allowance of up to 50 percent is provided as long as the building is equipped throughout with an automatic sprinkler system. The reference to Section 903.3.1 allows the use of a NFPA 13 or 13R system as applicable. This allowance is likely provided as it is more realistic to what is typically found in dormitories and is more easily managed from an enforcement standpoint. The sprinklers provide the necessary protection to allow such an increase.

Exception 3 clarifies that, in Group B and M occupancies, fabric partitions suspended from the ceiling but not physically contacting the floor should be treated as decorative materials similar to curtains or draperies and comply with the flame propagation performance heat release criteria of Section 807.4. These partitions are also permitted to be unlimited. This is with or without the use of sprinklers within a building.

Exception 4 provides specific clarification that the 10 percent limit does not apply to combustible materials that are used specifically as window coverings.

807.3 Acceptance criteria and reports. Where required to exhibit improved fire performance, curtains, draperies, fabric hangings and other similar combustible decorative materials suspended from walls or ceilings shall be tested by an *approved* agency and meet the flame propagation performance criteria of Test Method 1 or Test Method 2, as appropriate, of NFPA 701 or exhibit a maximum rate of heat release of 100 kW when tested in accordance with NFPA 289, using the 20 kW ignition source. Reports of test results shall be prepared in accordance with the test method used and furnished to the *fire code official* upon request.

C One of the standard test methods used to evaluate the ability of a material to propagate flame is NFPA 701, which contains two test methods. Test 1 is less severe than Test 2 and uses smaller specimens. Test 1 is intended for lighter weight materials and single-layer fabrics. The limit is a linear density of 700 g/m^2 (21 oz/yd^2). Test 2 is intended for higher density materials and multilayered fabrics. It also applies to vinyl-coated fabric blackout linings and lined draperies using any density, because they have been shown to give misleading results using Test 1. NFPA 701 sets out the types of materials, including fabrics that should be tested using each method.

Essentially, NFPA 701 provides a mechanism to distinguish between materials that allow flames to spread quickly and those that do not when using a small fire exposure.

Materials tested only to NFPA 701 are not permitted for use as interior finish materials, but instead are generally used as shades, swags, curtains and other similar materials. These tests are used to determine whether materials propagate flame beyond the area exposed to the ignition source. They are not intended to indicate whether the material tested will resist the propagation of flame under fire exposures more extreme than the test conditions.

It should be noted that the historic "small-scale test" is no longer accepted. The other standard test method that can be used is NFPA 289. This test method quantifies the contribution of materials to heat and smoke release when subjected to different ignition sources. This test method also determines the potential for growth of a fire and the fire-spread of a given material when exposed to an ignition source in a controlled environment. See the commentary to Section 807.5.1.1 for a more detailed discussion of this test method.

807.4 Artificial decorative vegetation. Artificial decorative vegetation shall comply with this section and the requirements of Sections 806.2 and 806.3. Natural decorative vegetation shall comply with Section 806.

Exception: Testing of artificial vegetation is not required in Group I-1; Group I-2, Condition 1; Group R-2; Group R-3; or Group R-4 occupancies equipped throughout with an *approved automatic sprinkler system* installed in accordance with Section 903.3.1, where such artificial vegetation complies with the following:

1. Wreaths and other decorative items on doors shall not obstruct the door operation and shall not exceed 50 percent of the surface area of the door.
2. Decorative artificial vegetation shall be limited to not more than 30 percent of the wall area to which it is attached.
3. Decorative artificial vegetation not on doors or walls shall not exceed 3 feet (914 mm) in any dimension.

C This section addresses artificial decorative vegetation, which generally contains a high concentration of plastic material in the form of plastic artificial leaves. The goal, therefore, is to reduce initial ignitability by treating artificial vegetation with flame retardants to improve fire performance or to use materials that have inherently better fire performance. The exception for the specific occupancies listed provides the limited conditions where it can be used without having to meet the testing requirements in buildings that are fully protected with an automatic sprinkler system.

807.4.1 Flammability. Artificial decorative vegetation shall meet the flame propagation performance criteria of Test Method 1 or Test Method 2, as appropriate, of NFPA 701. Meeting such criteria shall be documented and certified by the manufacturer in an *approved* manner. Alternatively, the artificial decorative vegetation shall be tested in accordance with NFPA 289, using the 20 kW ignition source, and shall have a maximum heat release rate of 100 kW.

C Artificial decorative vegetation is required by this section to be tested either to the flame propagation performance criteria of NFPA 701 or in accordance with NFPA 289 using the 20 kW ignition source and with a maximum allowed heat release rate of 100 kW.

NFPA 289 is a furniture calorimeter heat release fire test specifically developed for these types of products. NFPA 289 uses a propane gas burner as the ignition source. The 20 kW gas burner ignition source in NFPA 289 was specifically designed with the intent of being a substitute for UL 1975 and would be suitable as an ignition source for Christmas trees.

As discussed extensively in Section 806.1, there are substantial fire safety requirements for natural Christmas trees; however, testing in accordance with NFPA 701 was the only option until this requirement was added.

807.4.2 Electrical fixtures and wiring on artificial vegetation. The use of unlisted electrical wiring and lighting on artificial decorative vegetation shall be prohibited. The use of electrical wiring and lighting on artificial trees constructed entirely of metal shall be prohibited.

C Decorations are quite often used year after year on Christmas trees. Some of this decor consists of lights that are not specifically listed and can fail and become very hot, potentially posing an ignition source. Additionally, this wiring may arc and create an ignition source. In fact, it has been found that many Christmas tree fires have actually been started by decorative lights on the tree. Therefore, this section simply prohibits the use of unlisted wiring or lighting (such lights and ornaments are listed in accordance with UL 588). Due to the shock and ignition potential resulting from a short or direct contact of part of the tree with one of the light sockets, electrical wiring and lighting is not allowed on metal trees.

807.5 Occupancy-based requirements. Occupancies shall comply with Sections 807.5.1 through 807.5.6.

C The provisions of Sections 807.5.1 through 807.5.6 address the vulnerability of occupants by addressing hazards likely found in certain occupancies. Group A occupancies are generally densely populated with occupants being unfamiliar with their surroundings. Such occupancies are also more likely to contain a large amount of decorative materials, such as stage scenery or exhibit booths utilizing foam plastics. Group E, I and R-2 dormitories have the potential to deal with a large amount of decorative materials and storage. Occupants in Group E, I and R-2 dormitories are also considered more vulnerable than other occupancies, such as Group B or M, because of occupant age or infirmity and occupancies such as dormitories where the occupants sometimes place themselves at increased risks. Group I-3, as will be discussed, should not have any combustible decorative materials. It is important to stress that this section specifically addresses materials that do not meet the requirements of NFPA 701, as is required in Section 807.3.

807.5.1 Group A. In Group A occupancies, the requirements in Sections 807.5.1.1 through 807.5.1.4 shall apply.

C The requirements in Sections 803.5.1.1 through 803.5.1.4 are specific to Group A occupancies. These particular requirements are fairly specific to activities and uses that occur in Group A occupancies, such as trade shows and movie theaters. Generally, fire hazards are moderate in Group A occupancies. The concerns are more closely related to the high occupant density and the occupants' lack of familiarity with the building.

807.5.1.1 Foam plastics. Exposed foam plastic materials and unprotected materials containing foam plastic used for decorative purposes or stage scenery or exhibit booths shall have a maximum heat release rate of 100 kW when tested in accordance with UL 1975, or when tested in accordance with NFPA 289 using the 20 kW ignition source.

Exceptions:

1. Individual foam plastic items or items containing foam plastic where the foam plastic does not exceed 1 pound (0.45 kg) in weight.
2. Cellular or foam plastic shall be allowed for trim in accordance with Section 804.2.

C As discussed in Section 803.8, foam plastics can produce misleading results when tested to ASTM E84. In some instances, they can burn vigorously and at high rates of heat release. Group A occupancies tend to contain materials for stages, movie theaters and exhibit halls. These uses will likely have combustible scenery or exhibit booths that contain foam plastics. Therefore, because the occupants are unlikely to be familiar with the premises, and the hazards presented by these combustibles can be high, a heat release rate limit of 100 kW, when tested in accordance with UL 1975 or NFPA 289, is placed on foam plastic materials used in Group A occupancies.

In the past, restrictions on the combustibility of materials were based on the heat content of the materials instead of the rate at which the heat content is released. The problem with using heat content (which can be equated to "potential energy" of the material) as the limitation criterion is that it does not provide a good understanding of the rate at which the "potential energy" is released. The rate at which heat is released is a more important characteristic of fire and is a measure of the intensity of the fire. UL 1975 and NFPA 289 are furniture calorimeter types of tests that assess the actual heat release rate based on the exposure of the foam plastics to a specific heat input (340-gram wood crib for UL 1975 and 20kW propane gas burner ignition source for NFPA 289).

Some alternatives to testing are provided in the exceptions, which primarily focus on limiting the amount of combustibles. The exceptions give two prescriptive approaches with no direct relationship to the actual combustibility of the foam plastic except to limit the amount used. Exception 1 is related to plastic display items, such as a small statue. Exception 2 relates specifically to decorative trim and directs the code user to the requirements for that material found in Section 804.2 (see commentary, Section 804.2).

807.5.1.2 Motion picture screens. The screens on which motion pictures are projected in new and existing buildings of Group A shall either meet the flame propagation performance criteria of Test Method 1 or Test Method 2, as appropriate, of NFPA 701 or shall comply with the requirements for a Class B interior finish in accordance with Section 803 of the *International Building Code.*

C Movie screens are not considered interior finish and, therefore, would not be addressed by Chapter 8 of the IBC. These screens consist of a fabric base covered with a thin coating impregnated with reflective glass beads. Movie screens are generally fairly large and typically take up most of the front wall of a movie theater; therefore, the flame spread characteristics must be addressed. Because movie theaters tend to be densely occupied, a material with a high rate of flame spread can pose a significant hazard. Screens can comply in one of two ways. The screen must meet the applicable flame propagation performance criteria set out in NFPA 701 Test Method 1 or 2 (which focus on combustibles such as curtains, shades and window treatments, but are not appropriate for wall coverings), or qualify as a Class B interior finish material in accordance with ASTM E84. NFPA 286, which is also found in Section 803, would be a viable alternative for testing. It should be noted that the historic "small-scale test" in NFPA 701 is no longer accepted.

807.5.1.3 Wood use in places of religious worship. In places of religious worship, wood used for ornamental purposes, trusses, paneling or chancel furnishing shall not be limited.

C Places of religious worship are generally open spaces. The occupants, because of the nature of the activities in these assemblies, tend to be orderly. For this reason, wood, which is a combustible material, is allowed extensively as a finish material without restriction.

807.5.1.4 Pyroxylin plastic. Imitation leather or other material consisting of or coated with a pyroxylin or similarly hazardous base shall not be used.

C Use of pyroxylin plastics, also known as cellulose nitrate plastics, in Group A occupancies as imitation leather or other materials is strictly prohibited because of the normally high occupant loads of such occupancies. This type of material is very hazardous because it will begin decomposition at temperatures starting at 300°F (149°C) and has the potential to develop explosive atmospheres with high heat emission once it ignites. Because cellulose nitrate tends to become somewhat unstable and is easily ignitable, its use has generally declined. Specifically, use of cellulose nitrate for motion picture film was discontinued in 1951.

807.5.2 Group E. Group E occupancies shall comply with Sections 807.5.2.1 through 807.5.2.3.

C Group E occupancies are occupancies used by six or more persons for educational purposes through the 12th grade or buildings and structures used for educational, supervision or personal care services for more than five children over the age of 2 1/2 years. This section regulates the amount of combustibles within corridors, lobbies and classrooms by regulating clothing, personal effects and artwork.

807.5.2.1 Storage in corridors and lobbies. Clothing and personal effects shall not be stored in *corridors* and lobbies.

Exceptions:

1. *Corridors* protected by an *approved automatic sprinkler system* installed in accordance with Section 903.3.1.1.
2. *Corridors* protected by an *approved fire alarm system* installed in accordance with Section 907.
3. Storage in metal lockers, provided the minimum required egress width is maintained.

C Materials, such as clothing, other personal effects and artwork, have the potential for creating a fire hazard within the main path of egress travel. This section allows the storage of clothing and other personal effects within these areas only if corridors and lobbies contain one of the following features:

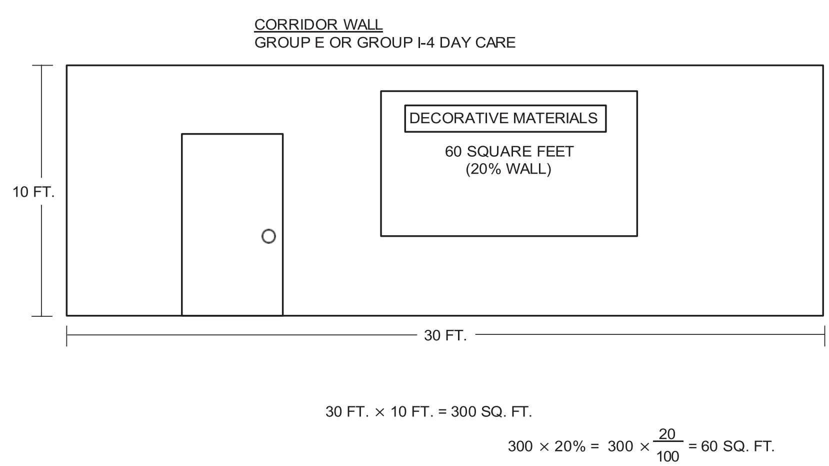
- A sprinkler system.
- A fire alarm system.
- Metal lockers for storage.

The sprinkler system must meet the requirements of NFPA 13. The fire alarm system must be approved by the fire code official. The fire alarm system is intended to specifically focus on the contents of the corridors, would typically be a smoke detection system and is not intended to be required throughout.

807.5.2.2 Artwork in corridors. Artwork and teaching materials shall be limited on the walls of *corridors* to not more than 20 percent of the wall area.

C Educational occupancies tend to display various artwork and related educational materials on the walls of classrooms and corridors. This section limits the potential combustibility levels of artwork in critical areas of the means of egress system; therefore, decorations or artwork can cover no more than 20 percent of the corridor walls. See Commentary Figure 807.5.2.2 for representation of the 20 percent coverage area.

Commentary Figure 807.5.2.2—Artwork Coverage Area Example



807.5.2.3 Artwork in classrooms. Artwork and teaching materials shall be limited on walls of classrooms to not more than 50 percent of the specific wall area to which they are attached.

C This is consistent with requirements for restrictions on artwork in the corridors. The limit is less restrictive at 50 percent versus the 20 percent that is permitted for corridors. This relates to the fact that the corridors make up the more critical portion of the exit access for the building.

807.5.3 Groups I-1 and I-2. In Group I-1 and I-2 occupancies, combustible *decorative materials* shall comply with Sections 807.5.3.1 through 807.5.3.4.

C Sections 807.5.3.1 through 807.5.3.4 provide a series of requirements for care facilities where the occupants are at increased level of risk due to their physical and cognitive limitations. In Group I-1 and I-2 occupancies, the occupants are present 24 hours a day with varying levels of care depending on the type of occupancy. IBC Section 407.2.1 allows waiting or similar areas to be open to corridors. These types of spaces typically have magazines, bulletin boards with paper notices tacked to them, and other combustible items not treated with flame retardants or tested to NFPA 701. Allowing a specified percentage of untreated, combustible decorative materials in Group I-1 and I-2 buildings equipped throughout with an automatic sprinkler system will not exceed the "ordinary occupancy" classification outlined in NFPA 13, nor does it increase the fire loading above what is currently permitted. These requirements offer consistent language to aid enforcement and a guide to providers to determine compliance within their facilities. The intent of these requirements is to eliminate the haphazard and inconsistent application of these provisions in facilities nationwide.

807.5.3.1 Group I-1 and I-2 Condition 1 within units. In Group I-1 and Group I-2, Condition 1 occupancies, equipped throughout with an *approved automatic sprinkler system* installed in accordance with Section 903.3.1.1, within *sleeping units* and *dwelling units*, combustible decorative materials placed on walls shall be limited to not more than 50 percent of the wall area to which they are attached.

C This section is focused on the sleeping units and dwelling units within Group I-1, which are typically assisted living and similar occupancies, and Group I-2, Condition 1, which are nursing homes. Since the requirements are focused within the sleeping and dwelling unit, the requirements are more lenient than those associated with the main exit access. Similar to the artwork restrictions for classrooms, such spaces are permitted to have 50 percent of the wall area occupied by combustible decorative materials. It should be noted that Group I-2 occupancies require the retroactive installation of sprinklers in accordance with Section 1105.9. All new Group I occupancies require the installation of an automatic sprinkler system.

807.5.3.2 In Group I-1 and I-2, Condition 1 for areas other than within units. In Group I-1 and Group I-2, Condition 1 occupancies, equipped throughout with an *approved automatic sprinkler system* installed in accordance with Section 903.3.1.1, combustible decorative materials placed on walls in areas other than within *dwelling* and *sleeping units* shall be limited to not more than 30 percent of the wall area to which they are attached.

C This section focuses on the same occupancies as Section 807.5.3.1, but on areas outside the sleeping and dwelling units. The allowance in sprinklered buildings is 30 percent versus the 50 percent in Section 807.5.3.1. These areas tend to expose a greater percentage of the building occupants to the hazards of combustible decorative materials. It should be noted that Group I-2 occupancies require the retroactive installation of sprinklers in accordance with Section 1105.9. All new Group I occupancies require the installation of an automatic sprinkler system.

807.5.3.3 In Group I-2, Condition 2. In Group I-2, Condition 2 occupancies, equipped throughout with an *approved automatic sprinkler system* installed in accordance with Section 903.3.1.1, combustible decorative materials placed on walls shall be limited to not more than 30 percent of the wall area to which they are attached.

C This section is the same as Section 807.5.3.2, except that it is focused on hospitals—specifically, areas of the hospital where more occupants are exposed to combustible decorative materials. As noted in Sections 807.5.3.1 and 807.5.3.2, all Group I-2 occupancies require the retroactive installation of an automatic sprinkler system. Note that Section 1105.9 only requires sprinkler installation from the story containing the Group I-2 occupancy to the level of exit discharge.

807.5.3.4 Other areas in Groups I-1 and I-2. In Group I-1 and I-2 occupancies, in areas not equipped throughout with an *approved automatic sprinkler system*, combustible *decorative materials* shall be of such limited quantities that a hazard of fire development or spread is not present.

C This section addresses Group I-1 and I-2 occupancies. The focus is on areas of such buildings where an automatic sprinkler system does not provide coverage. As noted, Group I-2 occupancies require the retroactive installation of an automatic sprinkler system but Group I-1 occupancies do not. Although Group I-2 occupancies should be provided with sprinklers, there may be transitional periods where this is not the case. This requirement in general is nonspecific and generally gives authority to the fire code official to address situations that appear hazardous.

807.5.4 Group I-3. In Group I-3, combustible *decorative materials* are prohibited.

C Due to the nature of such occupancies, where occupants have a tendency to start fires, combustible decorative materials are prohibited.

807.5.5 Group I-4. Group I-4 occupancies shall comply with the requirements in Sections 807.5.5.1 through 807.5.5.3.

C A Group I-4 day care facility is an occupancy that cares for people of any age for less than 24 hours per day. Most often, these facilities are for small children not yet able to attend school, but also included are adult day care facilities. These occupancies will have features similar to those of Group E occupancies; therefore, the same restrictions on combustibles in corridors, lobbies and classrooms exist. The key issue here is that the occupants of Group I-4 facilities are often not capable of self-preservation without some level of assistance; therefore, combustibles must be kept to a minimum, especially in critical portions of the means of egress system.

807.5.5.1 Storage in corridors and lobbies. Clothing and personal effects shall not be stored in *corridors* and lobbies.

Exceptions:

1. *Corridors* protected by an *approved automatic sprinkler system* installed in accordance with Section 903.3.1.1.

2. *Corridors* protected by an *approved fire alarm system* installed in accordance with Section 907.
3. Storage in metal lockers, provided that the minimum required egress width is maintained.

C See the commentary to Section 807.5.2.1.

807.5.5.2 Artwork in corridors. Artwork and teaching materials shall be limited on walls of *corridors* to not more than 20 percent of the wall area.

C See the commentary to Section 807.5.2.2.

807.5.5.3 Artwork in classrooms. Artwork and teaching materials shall be limited on walls of classrooms to not more than 50 percent of the specific wall area to which they are attached.

C See the commentary to Section 807.5.2.3.

807.5.6 Dormitories in Group R-2. In Group R-2 dormitories, within *sleeping units* and *dwelling units*, the combustible decorative materials shall be of limited quantities such that a hazard of fire development or spread is not present.

C This section is consistent with the needs of facilities where occupants often take unnecessary risks. This provides a mechanism for the fire code official to limit the potential hazards in such sleeping units and dwelling units. Note that the exception to Section 807.3 allows up to 50 percent of the aggregate wall areas to contain decorative materials if an automatic sprinkler system is provided.

SECTION 808—FURNISHINGS OTHER THAN UPHOLSTERED FURNITURE AND MATTRESSES OR DECORATIVE MATERIALS IN NEW AND EXISTING BUILDINGS

808.1 Waste and linen containers in Group I-1, I-2 and I-3 occupancies and ambulatory care facilities. Waste and linen containers located in Group I-1, I-2 and I-3 occupancies and ambulatory care facilities shall comply with Section 304.3.6.

C This section provides the direct reference to Section 304.3.6 for the waste and linen container requirements in the specific occupancies listed. See also the commentary to Section 304.3.6.

808.2 Signs. Foam plastic signs that are not affixed to interior building surfaces shall have a maximum heat release rate of 150 kW when tested in accordance with UL 1975, or when tested in accordance with NFPA 289 using the 20-kW ignition source.

Exception: Where the aggregate area of foam plastic signs is less than 10 percent of the floor area or wall area of the room or space in which the signs are located, whichever is less, subject to the approval of the *fire code official*.

C This section correlates with Section 402.6.4.5 of the IBC, which requires a limit on the combustibility of signs in covered mall buildings, though it should be noted that the requirements of this code are not specific to such buildings. The requirement of this section is that, when a foam plastic sign is tested in accordance with UL 1975 or NFPA 289 (see commentary, Section 807.5.1.1), a maximum heat release rate of 150 kW is allowed.

Note that smoke development is not part of the acceptance criteria.

The exception recognizes that a relatively small aggregate surface coverage of a room or space by these signs (i.e., when the aggregate area of foam plastic signs constitutes the lesser of 10 percent of the floor or wall area of the room or space in which the signs are located) does not present a sufficient enough hazard to require compliance with the section. However, specific approval by the fire code official is required.

808.3 Combustible lockers. Where lockers constructed of combustible materials are used, the lockers shall be considered to be interior finish and shall comply with Section 803.

Exception: Lockers constructed entirely of wood and noncombustible materials shall be permitted to be used wherever interior finish materials are required to meet a Class C classification in accordance with Section 803.1.2.

C Traditionally, lockers in schools (high schools, middle schools, universities), clubs, swimming pools and gymnasiums were constructed of steel. In recent years, the use of lockers constructed of combustible materials has become prevalent. These lockers typically line an entire wall (for example, a corridor in a school) and are not regulated by the code. Lockers are not usually considered interior finish. The only other materials regulated by the code at present are interior trim, upholstered furniture, decorations, decorative vegetation, wastebaskets, linen containers and signs. Lockers do not fall into any of those categories.

Combustible lockers can present a significant fire load and, if ignited, are likely to spread fire the same way that interior finish materials spread fire. They are considered interior finish materials and regulated like all other interior finish materials for any occupancy. Plastic lockers have the benefit of being more immune to the effects of water from wet clothing and are generally very durable—but, with these benefits comes a fire hazard. The lockers by one manufacturer are constructed of 3/8-inch-thick (9.52 mm) solid plastic bodies and heavy duty 1/2-inch-thick (12.7 mm) doors. Typically, the "solid plastic" used is either high-density polyethylene or polypropylene. Polypropylene, as discussed in the commentary to Section 803.9, is a thermoplastic that gives off considerable energy and produces a pooling flammable liquid fire when it burns.

808.4 Play structures added to existing buildings. Where play structures that exceed 10 feet (3048 mm) in height or 150 square feet (14 m^2) in area are added inside an existing building, they shall comply with Section 424 of the *International Building Code*.

C This section references the IBC for play structures that are added to existing occupancies. The fire testing that led to its inclusion in the IBC shows that these structures can release very large amounts of heat when they burn and need to be regulated. IBC Section 424 already regulates these structures for new buildings at the time of construction of the entire building; however, play structures are often added into malls or restaurants after the certificate of occupancy has been granted. Note that as these structures age, they are often removed and replaced, and the IBC would normally not apply. Similarly, the *International Existing Building Code*® (IEBC®) would also not apply.

Bibliography

The following resource material was originally used in the preparation of the commentary for this chapter of the code:

DeLauter, L., J. Lee, G. Roadarmel and D.W. Stroup, *Scotch Pine Christmas Tree Fire Tests*. FR4010. Gaithersburg, MD: National Institute of Standards and Technology, US Department of Commerce, 1999.

Fisher, F. L., McCracken, W., and Williamson, R. B., “Room Fire Experiments of Textile Wall Coverings: A Final Report of All Materials Tested Between March 1985 and January 1986,” Final Report to American Textile Manufacturers' Institute, ES-7583, University of California Berkeley Service to Industry, Report No. 86-2, 031986.